Multiply the second equation by $\frac{6}{5}$:

$$\frac{6}{5}\left(\frac{5}{6}t\right) = \frac{6}{5}\left(\frac{100}{r+8}\right) \Rightarrow t = \frac{600}{5r+40}, \quad \text{so}$$

$$\frac{100}{r} = \frac{600}{5r+40}.$$

Cross-multiply:

$500r + 4000 = 600r \Rightarrow 100r = 4000 \Rightarrow r = 40.$

Henry drove at 40 miles per hour, and the trip took $100 \div 40 = 2.5$ hours = 150 minutes. (Had he driven at 48 miles per hour, the trip would have taken 125 minutes.)

10. C. Let x = Martin's weight in 1950. By 1980, he had gained 60 pounds (2 pounds per year for 30 years) and was 40% heavier:

$$60 = 0.40x \Rightarrow x = 60 \div 0.4 = 150.$$

In 1980, he weighed 210 pounds, and 15 years later, in 1995, he weighed 240:

$$\frac{210}{240} = \frac{7}{8} = 87.5\%.$$

11. (60) Let x = the greater, and y = the smaller, of the two numbers; then

$$(x + y) = 30 + (x - y) \Rightarrow y = 30 - y \Rightarrow$$
$$2y = 30 \Rightarrow y = 15;$$

and, since $xy = 900$, $x = 900 \div 15 = 60$.

12. (720) If x = number of shells in Phil's collection, then Fred has $.80x$. Since Phil has 80 more shells than Fred:

$$x = .80x + 80 \Rightarrow .20x = 80 \Rightarrow$$
$$x = 80 \div .20 = 400.$$

Phil has 400 and Fred has 320: a total of 720.

13. (30) Use TACTIC D1. Karen won $3x$ times and lost $2x$ times, and thus played a total of $5x$ games. Since she got \$5 every time she won, she received $\$5(3x) = \$15x$. Also, since she paid \$2 for each loss, she paid out $\$2(2x) = \$4x$. Therefore, her net winnings were $\$15x - \$4x = \$11x$, which you are told was \$66. Then, $11x = 66 \Rightarrow x = 6$, and so $5x = 30$.

14. (18) Since the average of all 10 players was 75, the total number of baskets made was $10 \times 75 = 750$. Also, since 8 of the players had an average of 79, they made a total of $8 \times 79 = 632$ points. The other 2 players, therefore, made $750 - 632 = 118$ baskets. The most baskets that the player with the highest number could have made was 100, so the player with the lowest number had to have made at least 18.

15. (50) You can avoid some messy algebra by working backwards. Put back the \$11 Stan took; then put back the extra \$1 that Dan took. There is now \$12, which means that, when Dan took his half, he took \$12. Put that back. Now there is \$24 in the box. Put back the extra \$1 that Ann took. The box now has \$25, so before Ann took her half, there was \$50. *Algebraic solution.* Assume that there were originally x dollars in the box. Ann took $\frac{1}{2}x + 1$, leaving $\frac{1}{2}x - 1$. Dan then took $\frac{1}{2}$ of that plus \$1 more; he took

$$\frac{1}{2}\left(\frac{1}{2}x - 1\right) + 1 = \frac{1}{4}x - \frac{1}{2} + 1 = \frac{1}{4}x + \frac{1}{2}.$$

Then Stan took \$11. Since together they took all x dollars:

$$x = \left(\frac{1}{2}x + 1\right) + \left(\frac{1}{4}x + \frac{1}{2}\right) + 11 = \frac{3}{4}x + 12\frac{1}{2}.$$

Therefore, $12\frac{1}{2} = \frac{1}{4}x \Rightarrow x = 50.$

Geometry

Although about 30% of the math questions on the SAT involve geometry, you need to know only a relatively small number of facts—far less than you would learn in a geometry course—and, of course, you need provide no proofs. The next six sections review all of the geometry that you need to know to do well on the SAT. Also, the material is presented exactly as it appears on the SAT, using the same vocabulary and notation, which may be slightly different from the terminology you have used in your math classes. There are plenty of sample multiple-choice and grid-in problems for you to solve, and they will show you exactly how these topics are treated on the SAT.

12-I LINES AND ANGLES

On the SAT, lines are usually referred to by lowercase letters, typically ℓ, m, and n. If P and Q are any points on line ℓ, we can also refer to ℓ as $\overleftrightarrow{PQ}$. In general, we have the following notations:

- $\overleftrightarrow{PQ}$ represents the ***line*** that goes through P and Q:

- $\overrightarrow{PQ}$ represents a ***ray***; it consists of point P and all the points on $\overleftrightarrow{PQ}$ that are on the same side of P as Q:

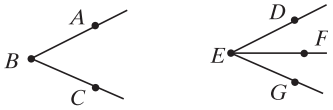
- $\overline{PQ}$ represents a ***line segment*** (often referred to simply as a ***segment***); it consists of points P and Q and all the points on $\overleftrightarrow{PQ}$ that are between them:

- PQ represents the ***length*** of segment $\overline{PQ}$.

If $\overline{AB}$ and $\overline{PQ}$ have the same length, we say that $\overline{AB}$ and $\overline{PQ}$ are ***congruent,*** and write $\overline{AB} \cong \overline{PQ}$. We can also write $AB = PQ$.

An ***angle*** is formed by the intersection of two line segments, rays, or lines. The point of intersection is called the ***vertex***.

An angle can be named by three points: a point on one side, the vertex, and a point on the other side. When there is no possible ambiguity, the angle can be named just by its vertex. For example, in the diagram below we can refer to the angle on the left as $\angle B$ or $\angle ABC$. To talk about $\angle E$, on the right, however, would be ambiguous; $\angle E$ might mean $\angle DEF$ or $\angle FEG$ or $\angle DEG$.



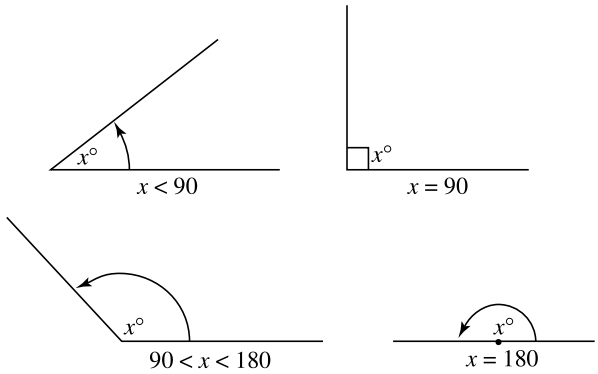
On the SAT, angles are always measured in degrees. The degree measure of $\angle ABC$ is represented by $m\angle ABC$. If $\angle P$ and $\angle Q$ have the same measure, we say that they are congruent and write $\angle P \cong \angle Q$. In the diagram below, $\angle A$ and $\angle B$ at the left are right angles. Therefore, $m\angle A = 90$ and $m\angle B = 90$, so $m\angle A = m\angle B$ and $\angle A \cong \angle B$. In equilateral triangle PQR, at the right, $m\angle P = m\angle Q = m\angle R = 60$, and $\angle P \cong \angle Q \cong \angle R$.

$\angle A \cong \angle B$ $\quad$ $\angle P \cong \angle Q \cong \angle R$

Key Fact I1

Angles are classified according to their degree measures.

- **An *acute* angle measures less than 90°.**
- **A *right* angle measures 90°.**
- **An *obtuse* angle measures more than 90° but less than 180°.**
- **A *straight* angle measures 180°.**

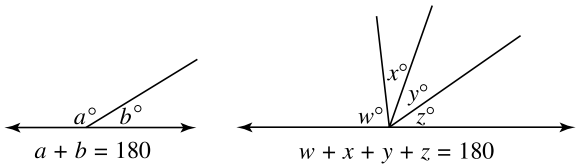


NOTE: A small square like the one in the second figure above *always* means that the angle is a right angle. On the SAT, if an angle has a square, it must be a 90° angle, *even if the figure has not been drawn to scale.*

Key Fact I2

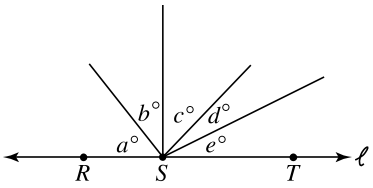
If two or more angles form a straight angle, the sum of their measures is 180°.

KEY FACT I2 is one of the facts provided in the "Reference Information" at the beginning of each math section.



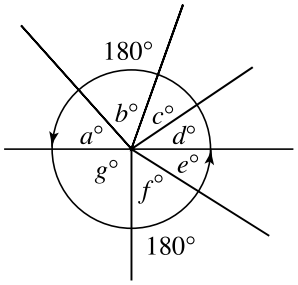
Example 1.

In the figure below, R, S, and T are all on line ℓ. What is the average (arithmetic mean) of a, b, c, d, and e?

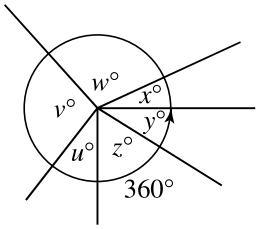


Solution. Since $\angle RST$ is a straight angle, by KEY FACT I2, the sum of a, b, c, d, and e is 180, and so their average is $\frac{180}{5} = \mathbf{36}$.

In the figure at the right, since $a + b + c + d = 180$ and $e + f + g = 180$, $a + b + c + d + e + f + g = 180 + 180 = 360$.



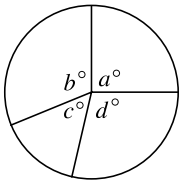
It is also true that $u + v + w + x + y + z = 360$, even though none of the angles forms a straight angle.



Key Fact I3

The sum of the measures of all the angles around a point is 360°.

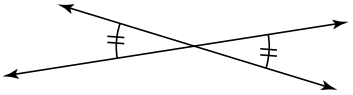
NOTE: This fact is particularly important when the point is the center of a circle, as will be seen in Section 12-L.



$a + b + c + d = 360$

KEY FACT I3 is one of the facts provided in the "Reference Information" at the beginning of each math section.

When two lines intersect, four angles are formed. The two angles in each pair of opposite angles are called ***vertical angles***.

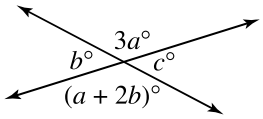


Key Fact I4

Vertical angles are congruent.

Example 2.

In the figure to the right, what is the value of a?



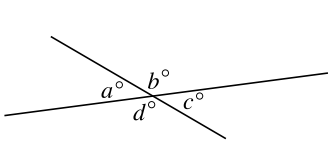
Solution. Because vertical angles are congruent:

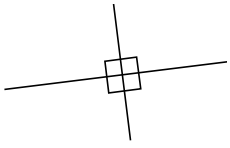
$$a + 2b = 3a \Rightarrow 2b = 2a \Rightarrow a = b.$$

For the same reason, $b = c$. Therefore, a, b, and c are all equal. Replace each b and c in the figure with a, and add:

$$a + a + 3a + a + 2a = 360 \Rightarrow 8a = 360 \Rightarrow a = \mathbf{45}.$$

Consider these vertical angles:





By KEY FACT I4, $a = c$ and $b = d$.

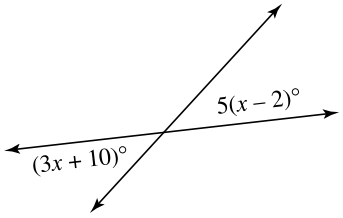
By KEY FACT I2, $a + b = 180$, $b + c = 180$, $c + d = 180$, and $a + d = 180$.

It follows that, if any of the four angles is a right angle, all the angles are right angles.

Example 3.

In the figure at the right, what is the value of x?

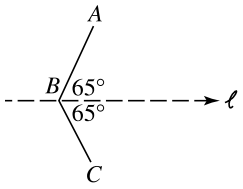
(A) 6 (B) 8
(C) 10 (D) 20
(E) 40

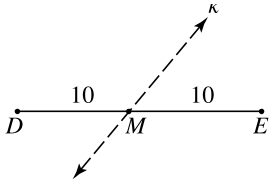


Solution. Since vertical angles are congruent:

$$3x + 10 = 5(x - 2) \Rightarrow 3x + 10 = 5x - 10 \Rightarrow$$
$$3x + 20 = 5x \Rightarrow 20 = 2x \Rightarrow x = \mathbf{10}\ \mathbf{(C)}.$$

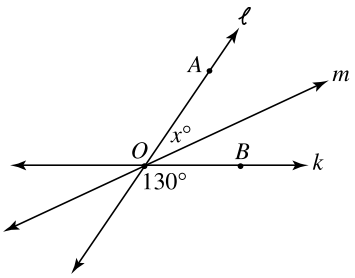
In the figures below, line ℓ divides $\angle ABC$ into two congruent angles, and line k divides line segment $\overline{DE}$ into two congruent segments. Line ℓ is said to ***bisect*** the angle, and line k ***bisects*** the line segment. Point M is called the ***midpoint*** of segment $\overline{DE}$.





Example 4.

In the figure at the right, lines k, ℓ, and m intersect at O. If line m bisects $\angle AOB$, what is the value of x?



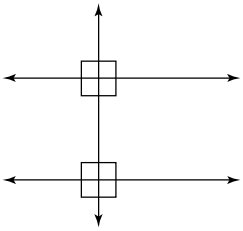
Solution. Here, $m\angle AOB + 130 = 180 \Rightarrow m\angle AOB = 50$; and since $\angle AOB$ is bisected, $x = \mathbf{25}$.

Two lines that intersect to form right angles are said to be ***perpendicular***.

Two lines that never intersect are said to be ***parallel***. Consequently, parallel lines form no angles. However, if a third line, called a ***transversal***, intersects a pair of parallel lines, eight angles are formed, and the relationships among these angles are very important.

Key Fact I5

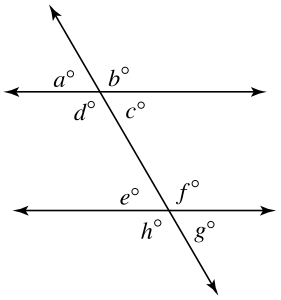
If a pair of parallel lines is cut by a transversal that is perpendicular to the parallel lines, all eight angles are right angles.



Key Fact I6

If a pair of parallel lines is cut by a transversal that is *not* perpendicular to the parallel lines:

- **Four of the angles are acute, and four are obtuse.**
- **All four acute angles are congruent: $a = c = e = g$.**
- **All four obtuse angles are congruent: $b = d = f = h$.**
- **The sum of any acute angle and any obtuse angle is 180°: for example, $d + e = 180$, $c + f = 180$, $b + g = 180$, ...**



Key Fact I7

If a pair of lines that are not parallel is cut by a transversal, *none* of the statements listed in KEY FACT I6 is true.

You *must* know KEY FACT I6—virtually every SAT has questions based on it. However, you do *not* need to know the special terms you learned in your geometry class for these pairs of angles; those terms are not used on the SAT.

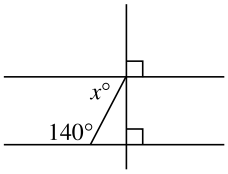
Key Fact I8

If a line is perpendicular to each of a pair of lines, then these lines are parallel.

Example 5.

What is the value of x in the figure at the right?

(A) 40
(B) 50
(C) 90
(D) 140
(E) It cannot be determined from the information given.



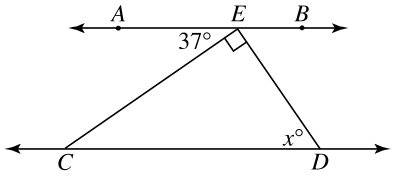
Note: Figure not drawn to scale

Solution. Despite the fact that the figure has not been drawn to scale, the little squares assure you that the vertical line is perpendicular to both of the horizontal ones, so these lines are parallel. Therefore, the sum of the 140° obtuse angle and the acute angle marked x° is 180°: $x + 140 = 180 \Rightarrow x =$ **40 (A).**

NOTE: If the two little squares indicating right angles were not in the figure, the answer would be E: "It cannot be determined from the information given." You are not told that the two lines that look parallel are actually parallel; and since the figure is not drawn to scale, you certainly cannot make that assumption. If the lines are not parallel, then 140 + x is *not* 180, and x cannot be determined.

Example 6.

In the figure below, $\overleftrightarrow{AB}$ is parallel to $\overleftrightarrow{CD}$. What is the value of x?



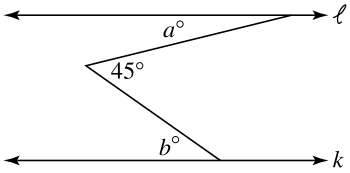
Solution. Let y be the measure of $\angle BED$. Then by KEY FACT I2:

$$37 + 90 + y = 180 \Rightarrow 127 + y = 180 \Rightarrow y = 53.$$

Since $\overleftrightarrow{AB}$ is parallel to $\overleftrightarrow{CD}$ by KEY FACT I6, $x = y \Rightarrow x =$ **53.**

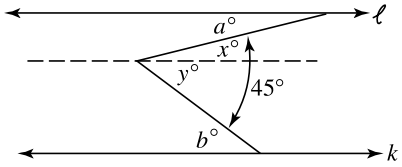
Example 7.

In the figure below, lines ℓ and k are parallel. What is the value of $a + b$?

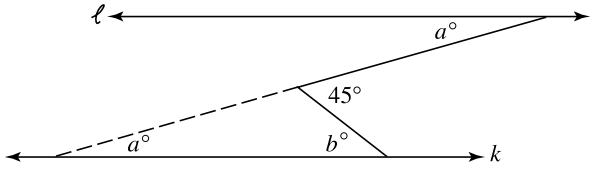


(A) 45 (B) 60 (C) 90 (D) 135
(E) It cannot be determined from the information given.

Solution. If you were asked for the value of either a or b, the answer would be E—neither one can be determined; but if you are clever, you can find the value of $a + b$. Draw a line parallel to ℓ and k through the vertex of the angle. Then, looking at the top two lines, you see that $a = x$, and looking at the bottom two lines, you have $b = y$. Therefore, $a + b = x + y =$ **45** (**A**).



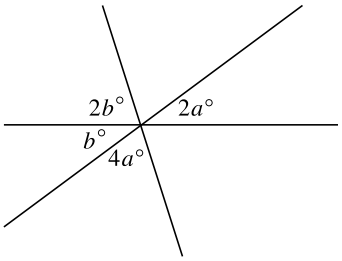
Alternative solution. Draw a different line and use a fact from Section 12-J on triangles. Extend one of the line segments to form a triangle. Since ℓ and k are parallel, the measure of the bottom angle in the triangle equals a. Now, use the fact that the sum of the measures of the three angles in a triangle is 180° or, even easier, that the given 45° angle is an external angle, and so is equal to the sum of a and b.



Exercises on Lines and Angles

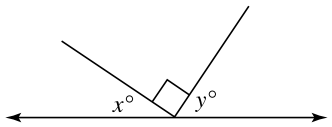
Multiple-Choice Questions

1. In the figure below, what is the value of b?



(A) 9 (B) 18 (C) 27 (D) 36 (E) 45

2. In the figure below, what is the value of x if $y{:}x = 3{:}2$?

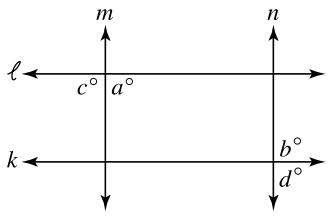


(A) 18 (B) 27 (C) 36 (D) 45 (E) 54

3. What is the measure of the angle formed by the minute and hour hands of a clock at 1:50?

(A) 90° (B) 95° (C) 105° (D) 115° (E) 120°

4. Concerning the figure below, if $a = b$, which of the following statements *must* be true?

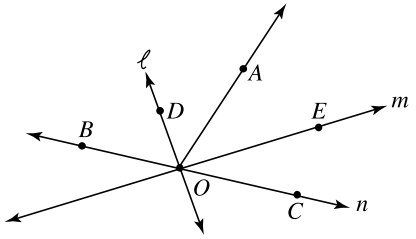


Note: Figure not drawn to scale

I. $c = d$.
II. ℓ and k are parallel.
III. m and ℓ are perpendicular.

(A) None (B) I only (C) I and II only
(D) I and III only (E) I, II, and III

5. In the figure below, B and C lie on line n, m bisects $\angle AOC$, and ℓ bisects $\angle AOB$. What is the measure of $\angle DOE$?

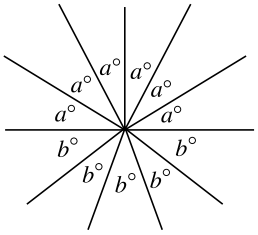


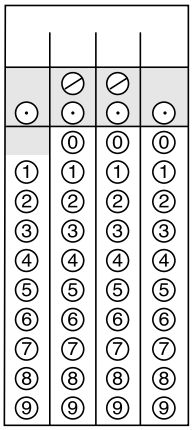
Note: Figure not drawn to scale

(A) 75 (B) 90 (C) 105 (D) 120
(E) It cannot be determined from the information given.

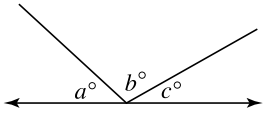
Grid-in Questions

6. In the figure below, what is the value of $\dfrac{b+a}{b-a}$?

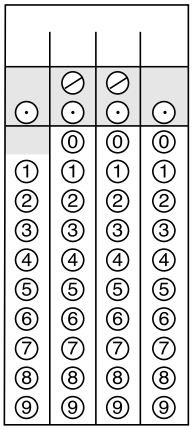




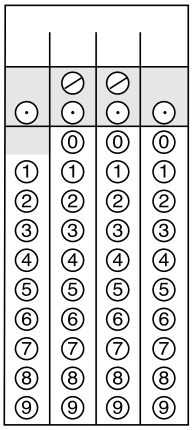
7. In the figure below, $a{:}b = 3{:}5$ and $c{:}b = 2{:}1$. What is the measure of the largest angle?



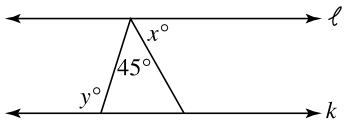
Note: Figure not drawn to scale

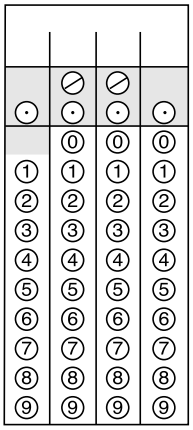


8. A, B, and C are points on a line, with B between A and C. Let M and N be the midpoints of $\overline{AB}$ and $\overline{BC}$, respectively. If $AB:BC = 3:1$, what is $AB:MN$?

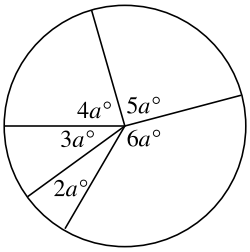


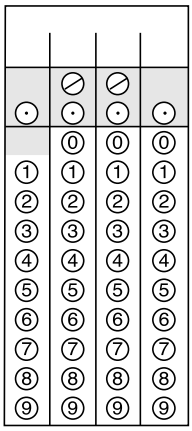
9. In the figure below, lines k and ℓ are parallel. What is the value of $y - x$?





10. In the figure below, what is the average (arithmetic mean) of the measures of the five angles?

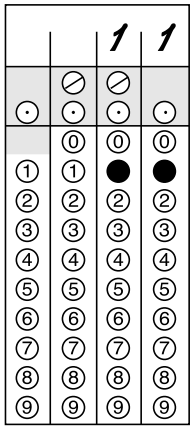




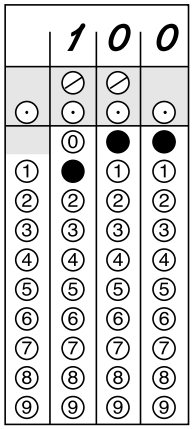
Answer Key

1. **D** 2. **C** 3. **D** 4. **B** 5. **B**

6.



7.



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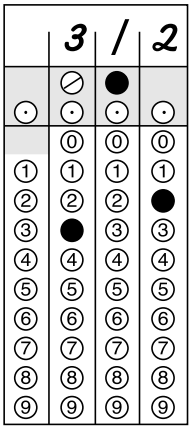
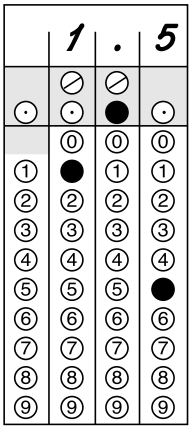


or



9.

		4	5
⊙	⊘ ⊙	⊘ ⊙	⊙
	⓪	⓪	⓪
①	①	①	①
②	②	②	②
③	③	③	③
④	④	●	④
⑤	⑤	⑤	●
⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨

10.

		7	2
⊙	⊘ ⊙	⊘ ⊙	⊙
	⓪	⓪	⓪
①	①	①	①
②	②	②	●
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥
⑦	⑦	●	⑦
⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨

Answer Explanations

1. D. Since vertical angles are equal, the two unmarked angles are $2b$ and $4a$. Also, since the sum of all six angles is 360°:

$$360 = 4a + 2b + 2a + 4a + 2b + b = 10a + 5b.$$

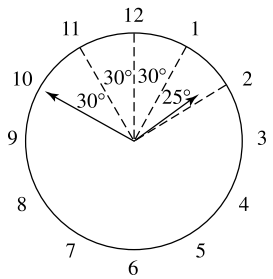
However, since vertical angles are equal, $b = 2a \Rightarrow 5b = 10a$. Hence:

$$360 = 10a + 5b = 10a + 10a = 20a \Rightarrow a = 18 \Rightarrow b = 36.$$

2. C. Since $x + y + 90 = 180$, then $x + y = 90$. Also, since $y{:}x = 3{:}2$, then $y = 3t$ and $x = 2t$. Therefore:

$$3t + 2t = 90 \Rightarrow 5t = 90 \Rightarrow t = 18 \Rightarrow x = 2(18) = 36.$$

3. D. For problems such as this, always draw a diagram. The measure of each of the 12 central angles from one number to the next on the clock is 30°. At 1:50 the minute hand is pointing at 10, and the hour hand has gone $\frac{50}{60} = \frac{5}{6}$ of the way from 1 to 2. Then, from 10 to 1 on the clock is 90°, and from 1 to the hour hand is $\frac{5}{6}(30°) = 25°$,



for a total of $90° + 25° = 115°$.

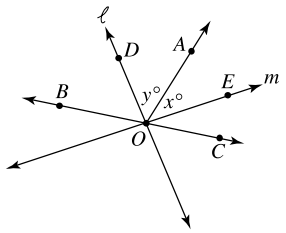
4. B. No conclusion can be drawn about the lines; they could form any angles whatsoever. (II and III are both false.) Statement I is true: $c = 180 - a = 180 - b = d$.

5. B. $x = \frac{1}{2}\text{m}\angle AOC$, and $y = \frac{1}{2}\text{m}\angle AOB$. Therefore,

$$x + y = \frac{1}{2}\text{m}\angle AOC + \frac{1}{2}\text{m}\angle AOB =$$

$$\frac{1}{2}(\text{m}\angle AOC + \text{m}\angle AOB) =$$

$$\frac{1}{2}(180) = 90.$$



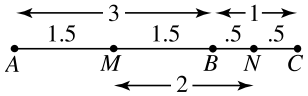
6. (11) From the diagram, you see that $6a = 180$, which implies that $a = 30$, and that $5b = 180$, which implies that $b = 36$. Therefore:

$$\frac{b+a}{b-a} = \frac{36+30}{36-30} = \frac{66}{6} = 11.$$

7. (100) Since $a{:}b = 3{:}5$, then $a = 3x$ and $b = 5x$; and since $c{:}b = 2{:}1$, then $c = 2b = 10x$. Then:

$$3x + 5x + 10x = 180 \Rightarrow 18x = 180 \Rightarrow x = 10 \Rightarrow c = 10x = 100.$$

8. $\left(\frac{3}{2} \text{ or } 1.5\right)$ If a diagram is not provided on a geometry question, draw one. From the figure below, you can see that $AB{:}MN = \frac{3}{2} = 1.5$.



9. (45) Since lines ℓ and k are parallel, the angle marked y in the given diagram and the sum of the angles marked x and 45 are equal: $y = x + 45 \Rightarrow y - x = 45$.

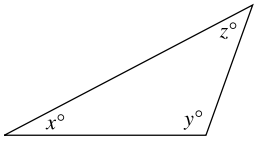
10. (72) The markings in the five angles are irrelevant. The sum of the measures of these angles is 360°, and $360 \div 5 = 72$. If you calculated the measure of each angle, you should have gotten 36, 54, 72, 90, and 108; but you wasted time.

12-J TRIANGLES

More geometry questions on the SAT pertain to triangles than to any other topic. To answer these questions correctly, you need to know several important facts about the angles and sides of triangles. The KEY FACTS in this section are extremely useful. Read them carefully, a few times if necessary, and *make sure you learn them all.*

Key Fact J1

In any triangle, the sum of the measures of the three angles is 180°.



$x + y + z = 180$.

KEY FACT J1 is one of the facts provided in the "Reference Information" at the beginning of each math section.

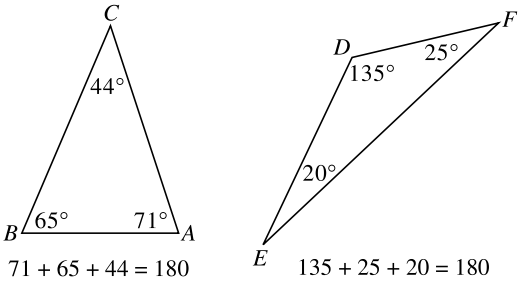
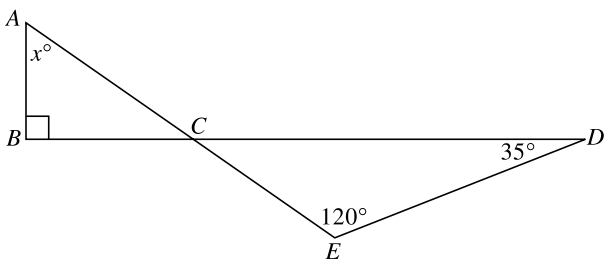


FIGURE 1

Figure 1 illustrates KEY FACT J1 for five different triangles, which will be discussed below.

Example 1.

In the figure below, what is the value of x?

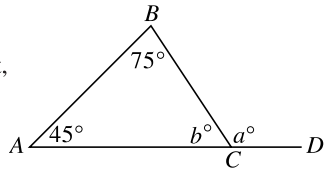


Solution. Use KEY FACT J1 twice: first, for $\triangle CDE$ and then for $\triangle ABC$.

- $m\angle DCE + 120 + 35 = 180 \Rightarrow m\angle DCE + 155 = 180 \Rightarrow m\angle DCE = 25$.
- Since vertical angles are congruent, $m\angle ACB = 25$ (see KEY FACT I4).
- $x + 90 + 25 = 180 \Rightarrow x + 115 = 180 \Rightarrow x =$ **65.**

Example 2.

In the figure at the right, what is the value of a?



Solution. First find the value of b:

$$180 = 45 + 75 + b = 120 + b \Rightarrow b = 60.$$

Then, $a + b = 180 \Rightarrow a = 180 - b = 180 - 60 =$ **120.**

In Example 2, $\angle BCD$, which is formed by one side of $\triangle ABC$ and the extension of another side, is called an ***exterior angle***. Note that, to find a, you did not have to first find b; you could just have added the other two angles: $a = 75 + 45 = 120$. This is a useful fact to remember.

Key Fact J2

The measure of an exterior angle of a triangle is equal to the sum of the measures of the two opposite interior angles.

Key Fact J3

In any triangle:

- **the longest side is opposite the largest angle;**
- **the shortest side is opposite the smallest angle;**
- **sides with the same length are opposite angles with the same measure.**

CAUTION: In KEY FACT J3 the condition "in any triangle" is crucial. If the angles are not in the same triangle, none of the conclusions holds. For example, in Figure 2 below AB, and DE are *not* equal even though each is opposite a 90° angle; and in Figure 3, QS is not the longest side even though it is opposite the largest angle.

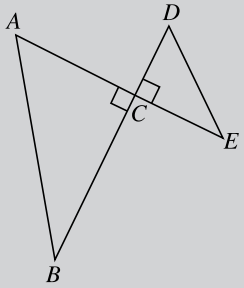


FIGURE 2

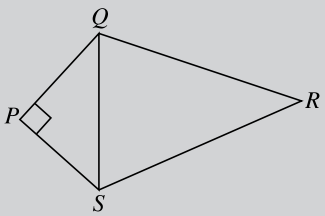


FIGURE 3

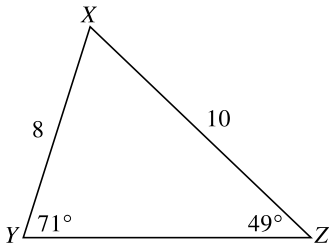
Consider triangles *ABC*, *JKL*, and *RST* in Figure 1.

- In $\triangle ABC$: *BC* is the longest side since it is opposite $\angle A$, the largest angle (71°). Similarly, *AB* is the shortest side since it is opposite $\angle C$, the smallest angle (44°). Therefore, $AB < AC < BC$.
- In $\triangle JKL$: Angles *J* and *L* have the same measure (45°), so $JK = KL$.
- In $\triangle RST$: Since all three angles have the same measure (60°), all three sides have the same length: $RS = ST = TR$.

Example 3.

In the figure at the right, which of the following statements concerning the length of side $\overline{YZ}$ is true?

(A) $YZ < 8$
(B) $YZ = 8$
(C) $8 < YZ < 10$
(D) $YZ = 10$
(E) $YZ > 10$



Solution.

- By KEY FACT J1, $m\angle X + 71 + 49 = 180 \Rightarrow m\angle X = 60$.
- Then *Y* is the largest angle, *Z* is the smallest, and *X* is in between.
- Therefore, by KEY FACT J3: $XY < YZ < XZ \Rightarrow 8 < YZ < 10$.
- The answer is **C**.

Classification of Triangles

Name	Lengths of Sides	Measures of Angles	Examples from Figure 1
Scalene	all 3 different	all 3 different	*ABC*, *DEF*, *GHI*
Isosceles	2 the same	2 the same	*JKL*
Equilateral	all 3 the same	all 3 the same	*RST*

Acute triangles are triangles such as *ABC* and *RST*, in which all three angles are acute. An acute triangle can be scalene, isosceles, or equilateral.

Obtuse triangles are triangles such as *DEF*, in which one angle is obtuse and two are acute. An obtuse triangle can be scalene or isosceles.

Right triangles are triangles such as *GHI* and *JKL*, which have one right angle and two acute ones. A right triangle can be scalene or isosceles. The side opposite the 90° angle is called the ***hypotenuse***, and by KEY FACT J3 it is the longest side. The other two sides are called the ***legs***.

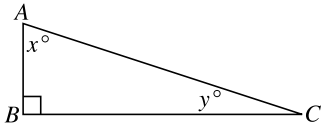
If *x* and *y* are the measures of the acute angles of a right triangle, then by KEY FACT J1: $90 + x + y = 180$, and so $x + y = 90$.

Key Fact J4

In any right triangle, the sum of the measures of the two acute angles is 90°.

Example 4.

In the figure below, what is the average (arithmetic mean) of *x* and *y*?



Solution. Since the diagram indicates that $\triangle ABC$ is a right triangle, then, by KEY FACT J4, $x + y = 90$. Therefore, the average of *x* and $y = \dfrac{x+y}{2} = \dfrac{90}{2} = \mathbf{45}$.

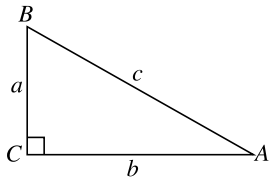
The most important facts concerning right triangles are the **Pythagorean theorem** and its converse, which are given in KEY FACTJ5 and repeated as the first line of KEY FACT J6.

The Pythagorean theorem is one of the facts provided in the "Reference Information" at the beginning of each math section.

Key Fact J5

Let *a*, *b*, and *c* be the sides of $\triangle ABC$, with $a \le b \le c$.

- **If $\triangle ABC$ is a right triangle, $a^2 + b^2 = c^2$;**
- **If $a^2 + b^2 = c^2$, then $\triangle ABC$ is a right triangle.**

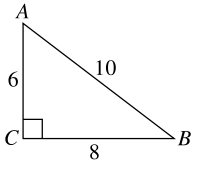


$a^2 + b^2 = c^2$

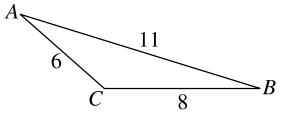
Key Fact J6

Let *a*, *b*, and *c* be the sides of $\triangle ABC$, with $a \le b \le c$.

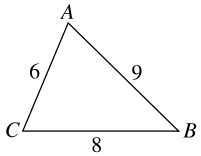
- **$a^2 + b^2 = c^2$ if and only if $\angle C$ is a right angle.**
- **$a^2 + b^2 < c^2$ if and only if $\angle C$ is obtuse.**
- **$a^2 + b^2 > c^2$ if and only if $\angle C$ is acute.**



$6^2 + 8^2 = 10^2$



$6^2 + 8^2 < 11^2$



$6^2 + 8^2 > 9^2$

Example 5.

Which of the following CANNOT be the lengths of the sides of a right triangle?

(A) 3, 4, 5 (B) 1, 1, $\sqrt{2}$ (C) 1, $\sqrt{3}$, 2

(D) $\sqrt{3}, \sqrt{4}, \sqrt{5}$ (E) 30, 40, 50

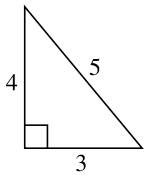
Solution. Just check the choices.

- (A): $3^2 + 4^2 = 9 + 16 = 25 = 5^2$ — These *are* the lengths of the sides of a right triangle.
- (B): $1^2 + 1^2 = 1 + 1 = 2 = (\sqrt{2})^2$ — These *are* the lengths of the sides of a right triangle.
- (C): $1^2 + (\sqrt{3})^2 = 1 + 3 = 4 = 2^2$ — These *are* the lengths of the sides of a right triangle.
- (D): $(\sqrt{3})^2 + (\sqrt{4})^2 = 3 + 4 = 7 \neq (\sqrt{5})^2$ — These *are not* the lengths of the sides of a right triangle.

Stop. The answer is **D**. There is no need to check choice E—but if you did, you would find that 30, 40, 50 *are* the lengths of the sides of a right triangle.

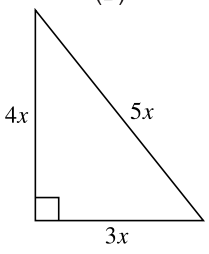
Below are the right triangles that appear most often on the SAT. You should recognize them immediately whenever they come up in questions. Carefully study each one, and memorize KEY FACTS J7–J11.

(A)



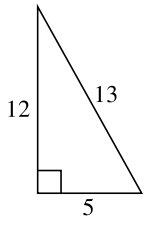
3, 4, 5

(B)



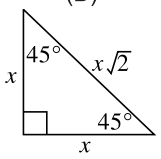
3x, 4x, 5x

(C)



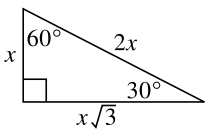
5, 12, 13

(D)



$x, x, x\sqrt{2}$

(E)



$x, x\sqrt{3}, 2x$

On the SAT, the most common right triangles whose sides are integers are the 3-4-5 right triangle (A) and its multiples (B).

Key Fact J7

For any positive number x, there is a right triangle whose sides are $3x$, $4x$, $5x$.

For example:

$x = 1$	3, 4, 5
$x = 2$	6, 8, 10
$x = 3$	9, 12, 15
$x = 4$	12, 16, 20
$x = 5$	15, 20, 25
$x = 10$	30, 40, 50
$x = 50$	150, 200, 250
$x = 100$	300, 400, 500

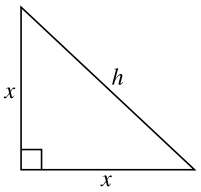
NOTE: KEY FACT J6 applies even if x is not an integer. For example:

$x = 0.5$ 1.5, 2, 2.5 $x = \pi$ $3\pi, 4\pi, 5\pi$

The only other right triangle with integer sides that you should recognize immediately is the one whose sides are 5, 12, 13 (C).

Let x = length of each leg, and h = length of the hypotenuse, of an isosceles right triangle (D). By the Pythagorean theorem,

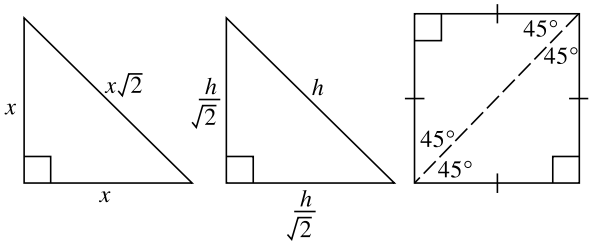
$$x^2 + x^2 = h^2 \Rightarrow 2x^2 = h^2 \Rightarrow h = \sqrt{2x^2} = x\sqrt{2}.$$



Key Fact J8

In a 45-45-90 right triangle, the sides are x, x, and $x\sqrt{2}$. Therefore:

- **By multiplying the length of a leg by $\sqrt{2}$, you get the hypotenuse.**
- **By dividing the hypotenuse by $\sqrt{2}$, you get the length of each leg.**



KEY FACT J8 is one of the facts provided in the "Reference Information" at the beginning of each math section.

Key Fact J9

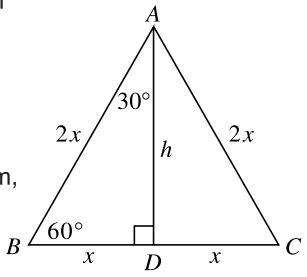
The diagonal of a square divides the square into two isosceles right triangles.

The last important right triangle is the one whose angles are 30°, 60°, and 90° (E).

Key Fact J10

An altitude divides an equilateral triangle into two 30-60-90 right triangles.

Let $2x$ be the length of each side of equilateral triangle ABC, in which altitude $\overline{AD}$ is drawn. Then $\triangle ADB$ is a 30-60-90 right triangle, and its sides are x, $2x$, and h. By the Pythagorean theorem, $x^2 + h^2 = (2x)^2 = 4x^2 \Rightarrow$ $h^2 = 3x^2 \Rightarrow$ $h = \sqrt{3x^2} = x\sqrt{3}$.

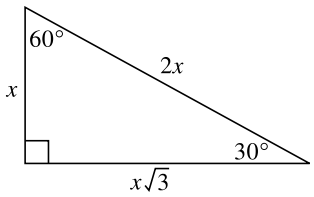


Key Fact J11

In a 30-60-90 right triangle the sides are x, $x\sqrt{3}$, and $2x$.

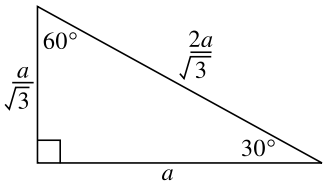
If you know the length of the shorter leg (x):

- multiply it by $\sqrt{3}$ to get the length of the longer leg;
- multiply it by 2 to get the length of the hypotenuse.



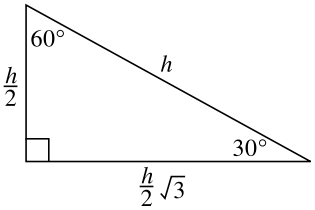
If you know the length of the longer leg (a):

- divide it by $\sqrt{3}$ to get the length of the shorter leg;
- multiply the shorter leg by 2 to get the length of the hypotenuse.



If you know the length of the hypotenuse (h):

- divide it by 2 to get the length of the shorter leg;
- multiply the shorter leg by $\sqrt{3}$ to get the length of the longer leg.



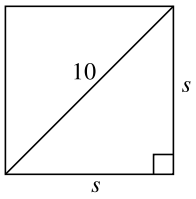
KEY FACT J11 is one of the facts provided in the "Reference Information" at the beginning of each math section.

Example 6.

What is the area of a square whose diagonal is 10?

(A) 20 (B) 40 (C) 50 (D) 100 (E) 200

Solution. Draw a diagonal in a square of side s, creating a 45-45-90 right triangle. By KEY FACT J8:



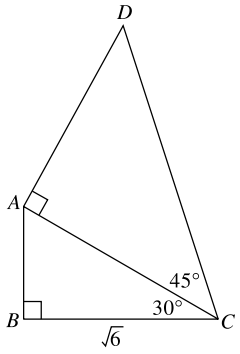
$$s = \frac{10}{\sqrt{2}} \quad \text{and} \quad A = s^2 = \left(\frac{10}{\sqrt{2}}\right)^2 = \frac{100}{2} = \mathbf{50}.$$

The answer is **C**.

[KEY FACT K8 gives the formula for the area of a square based on this example: $A = \frac{d^2}{2}$, where d is the length of a diagonal.]

Example 7.

In the diagram at the right, if $BC = \sqrt{6}$, what is the value of CD?



Solution. $\triangle ABC$ and $\triangle DAC$ are 30-60-90 and 45-45-90 right triangles, respectively. Use KEY FACTS J11 and J8.

- Divide $\overline{BC}$, the length of the longer leg, by $\sqrt{3}$ to get $\overline{AB}$, the length of the shorter leg: $\frac{\sqrt{6}}{\sqrt{3}} = \sqrt{2}$.
- Multiply $\overline{AB}$ by 2 to get the length of the hypotenuse: $AC = 2\sqrt{2}$.
- Since $\overline{AC}$ is also a leg of isosceles right triangle DAC, to get the length of hypotenuse $\overline{CD}$, multiply AC by $\sqrt{2}$: $CD = 2\sqrt{2} \times \sqrt{2} = 2 \times 2 = \mathbf{4}$.

Helpful Hint

If you know some elementary trigonometry, you could use the sine, cosine, and tangent ratios to solve questions involving 30-60-90 triangles and 45-45-90 triangles. But YOU SHOULDN'T. Rather, you should use KEY FACTS J8 and J11, just as was done in Example 7. You should know these facts by heart; but in case you forget, they are given to you in the Reference Information on the first page of each math section.

To solve Example 7 using trigonometry, you must first decide which trigonometric ratios to use, then write down the formulas, then manipulate them, and finally use your calculator to evaluate the answer.

The solution might look like this:

$$\cos\ 30° = \frac{\sqrt{6}}{AC} \Rightarrow (AC)\cos\ 30° = \sqrt{6} \Rightarrow AC = \frac{\sqrt{6}}{\cos\ 30°}.$$

Then on your calculator evaluate $\sqrt{6} \div \cos 30°$ = 2.18284... .

Don't round off. Leave the entire answer on your calculator screen.

Then write

$$\cos\ 45° = \frac{AC}{CD} \Rightarrow (CD)\cos\ 45° = AC \Rightarrow CD = \frac{AC}{\cos\ 45°}.$$

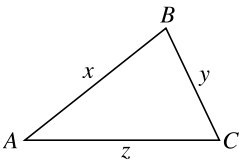
Now, on your calculator divide the number still on your screen by cos 45°. The answer is **4.**

Using trigonometry to solve this problem takes more time, not less, than the original solution, and the likelihood that you will make a mistake is far greater. Therefore, use nontrigonometric methods on all questions of this type.

Key Fact J12 (Triangle Inequality)

The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

The best way to remember this is to see that, in $\triangle ABC$, $x + y$, the length of the path from A to C through B, is greater than z, the length of the direct path from A to C.



$x + y > z$

NOTE: If you subtract x from each side of $x + y > z$, you see that $z - x < y$.

Key Fact J13

The difference between the lengths of any two sides of a triangle is less than the length of the third side.

Example 8.

If the lengths of two sides of a triangle are 6 and 7, which of the following could be the length of the third side?

I. 1 II. 5 III. 15

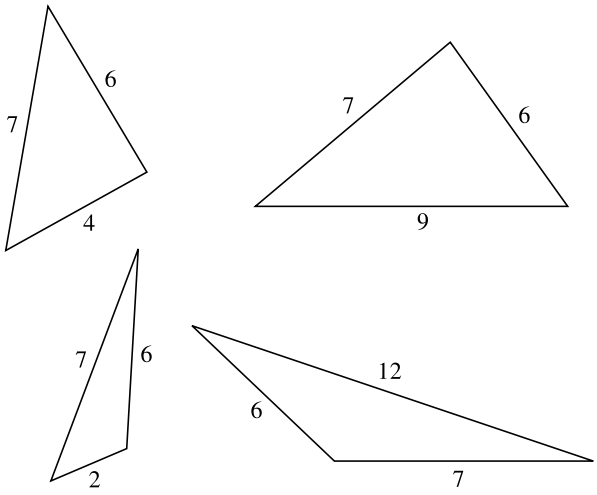
(A) None (B) I only (C) II only (D) I and II only (E) I, II, and III

Solution. Use KEY FACTS J12 and J13.

- The length of the third side must be *less* than 6 + 7 = 13. (III is false.)
- The length of the third side must be *greater* than 7 – 6 = 1. (I is false.)
- *Any* number between 1 and 13 could be the length of the third side. (II is true.)

The answer is **C.**

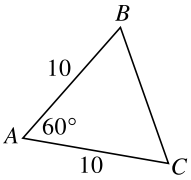
The following diagram illustrates several triangles two of whose sides have lengths of 6 and 7.



On the SAT, two other terms that appear regularly in triangle problems are ***perimeter*** and ***area*** (see Section 12-K).

Example 9.

In the figure at the right, what is the perimeter of $\triangle ABC$?



Solution. First, use KEY FACTS J3 and J1 to find the measures of the angles.

- Since $AB = AC$, $m\angle B = m\angle C$. Represent each measure by x.
- Then, $x + x + 60 = 180 \Rightarrow 2x = 120 \Rightarrow x = 60$.
- Since the measure of each angle of $\triangle ABC$ is 60°, the triangle is equilateral.
- Then, $BC = 10$, and the perimeter is 10 + 10 + 10 = **30.**

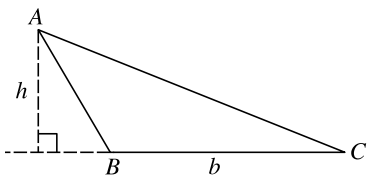
Key Fact J14

The area of a triangle is given by $A = \frac{1}{2}bh$, where b = base and h = height.

KEY FACT J14 is one of the facts provided in the "Reference Information" at the beginning of each math section.

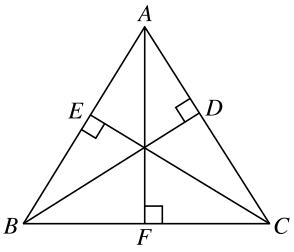
NOTE:

1. *Any* side of the triangle can be taken as the base.
2. The height is a line segment drawn perpendicular to the base from the opposite vertex.
3. In a right triangle, either leg can be the base and the other the height.
4. The height may be outside the triangle. [See the figure at the right.]



In the figure at the right:

- If $\overline{AC}$ is the base, $\overline{BD}$ is the height.
- If $\overline{AB}$ is the base, $\overline{CE}$ is the height.
- If $\overline{BC}$ is the base, $\overline{AF}$ is the height.



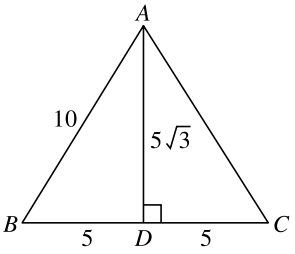
Example 10.

What is the area of an equilateral triangle whose sides are 10?

(A) 30 (B) $25\sqrt{3}$ (C) 50 (D) $50\sqrt{3}$ (E) 100

Solution. Draw an equilateral triangle and one of its altitudes.

- By KEY FACT J10, $\triangle ADB$ is a 30-60-90 right triangle.
- By KEY FACT J11, $BD = 5$ and $AD = 5\sqrt{3}$.
- The area of $\triangle ABC = \frac{1}{2}(10)(5\sqrt{3}) =$ **$25\sqrt{3}$** (**B**).

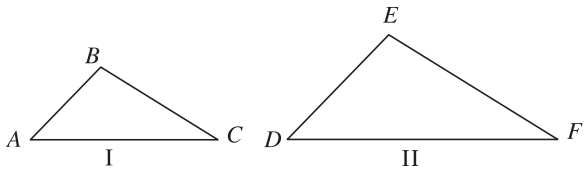


Replacing 10 by s in Example 10 yields a very useful result.

Key Fact J15

If A represents the area of an equilateral triangle with side s, then $A = \frac{s^2\sqrt{3}}{4}$.

Two triangles, such as I and II in the figure below, that have the same shape, but not necessarily the same size, are said to be ***similar***.



KEY FACT J16 makes this intuitive definition mathematically precise.

Key Fact J16

Two triangles are *similar* provided that the following two conditions are satisfied.

1. **The three angles in the first triangle are congruent to the three angles in the second triangle.**

   $m\angle A = m\angle D$, $m\angle B = m\angle E$, $m\angle C = m\angle F$.

2. **The lengths of the corresponding sides of the two triangles are in proportion:**

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}.$$

NOTE: Corresponding sides are sides opposite angles of the same measure.

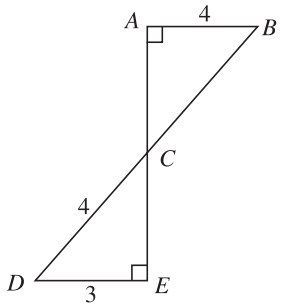
An important theorem in geometry states that, if condition 1 in KEY FACT J16 is satisfied, then condition 2 is automatically satisfied. Therefore, to show that two triangles are similar, it is sufficient to show that their angles have the same measure. Furthermore, if the measures of two angles of one triangle are equal to the measures of two angles of a second triangle, then the measures of the third angles are also equal. This is summarized in KEY FACT J17.

Key Fact J17

If the measures of two angles of one triangle are equal to the measures of two angles of a second triangle, the triangles are similar.

Example 11.

In the diagram at the right, what is BC?



Solution. Since vertical angles are congruent, $m\angle ECD = m\angle ACB$. Also, $m\angle A = m\angle E$ since both $\angle A$ and $\angle E$ are right angles. Then the measures of two angles of $\triangle CAB$ are equal to the measures of two angles of $\triangle CED$, and by KEY FACT J17, the two triangles are similar. Finally, by KEY FACT J16, corresponding sides are in proportion. Therefore:

$$\frac{DE}{AB} = \frac{DC}{BC} \Rightarrow \frac{3}{4} = \frac{4}{BC} \Rightarrow 3(BC) = 16 \Rightarrow BC = \mathbf{\frac{16}{3}}.$$

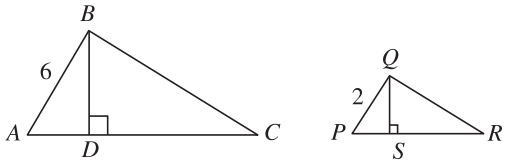
If two triangles are similar, the common ratio of their corresponding sides is called the ***ratio of similitude***.

Key Fact J18

If two triangles are similar, and if k is the ratio of similitude, then:

- **The ratio of all the linear measurements of the triangles is k.**
- **The ratio of the areas of the triangles is k^2.**

In the figure below, $\triangle ABC$ and $\triangle PQR$ are similar with $m\angle C = m\angle R$.



Then $\overline{AB}$ and $\overline{PQ}$ are corresponding sides, and the ratio of similitude is $\frac{6}{2} = 3$.

Therefore,

- All the sides are in the ratio of 3:1:

$$BC = \mathbf{3} \times QR, \qquad AC = \mathbf{3} \times PR.$$

- The altitudes are in the ratio of 3:1:

$$BD = \mathbf{3} \times QS.$$

- The perimeters are in the ratio of 3:1:

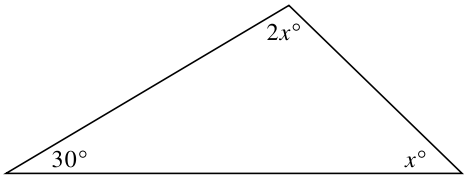
Perimeter of $\triangle ABC = \mathbf{3} \times$ (perimeter of $\triangle PQR$).

- The areas are in the ratio of 9:1:

Area of $\triangle ABC = \mathbf{9} \times$ (area of $\triangle PQR$).

Exercises on Triangles

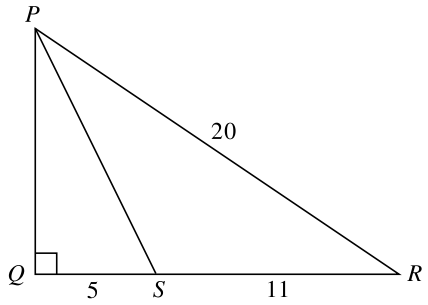
Multiple-Choice Questions



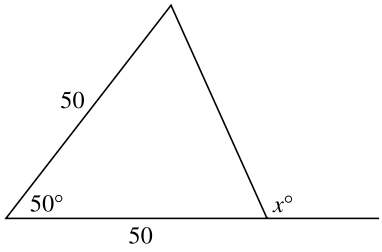
1. In the triangle above, what is the value of x?
   (A) 20 (B) 30 (C) 40 (D) 50 (E) 60

2. What is the area of an equilateral triangle whose altitude is 6?
   (A) 18 (B) $12\sqrt{3}$ (C) $18\sqrt{3}$ (D) 36
   (E) $24\sqrt{3}$

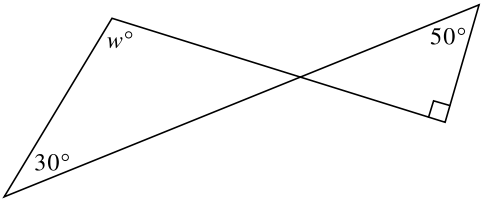
3. Two sides of a right triangle are 12 and 13. Which of the following could be the length of the third side?
   I. 5 II. 11 III. $\sqrt{313}$
   (A) I only (B) II only (C) I and II only
   (D) I and III only (E) I, II, and III



4. What is the value of PS in the triangle above?
   (A) $5\sqrt{2}$ (B) 10 (C) 11 (D) 13 (E) $12\sqrt{2}$

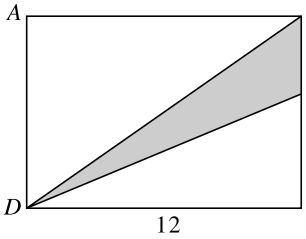


5. What is the value of x in the figure above?
   (A) 80 (B) 100 (C) 115 (D) 120 (E) 130



6. In the figure above, what is the value of w?
   (A) 100 (B) 110 (C) 120 (D) 130 (E) 140

Questions 7 and 8 refer to the following figure.



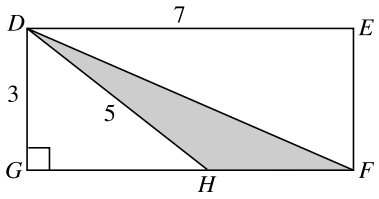
ABCD is a rectangle.

7. What is the area of $\triangle BED$?
   (A) 12 (B) 24 (C) 36 (D) 48 (E) 60

8. What is the perimeter of $\triangle BED$?

   (A) $19 + 5\sqrt{2}$ (B) 28 (C) $17 + \sqrt{185}$
   (D) 32 (E) 36

Questions 9 and 10 refer to the following figure.



DEFG is a rectangle.

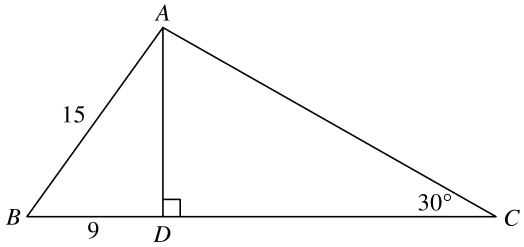
9. What is the area of $\triangle DFH$?

   (A) 3 (B) 4.5 (C) 6 (D) 7.5 (E) 10

10. What is the perimeter of $\triangle DFH$?

    (A) $8 + \sqrt{41}$ (B) $8 + \sqrt{58}$ (C) 16 (D) 17
    (E) 18

Questions 11 and 12 refer to the following figure.

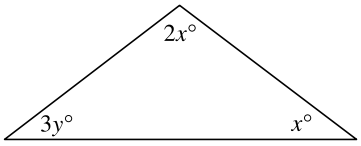


11. What is the perimeter of $\triangle ABC$?

    (A) 48
    (B) $48 + 12\sqrt{2}$
    (C) $48 + 12\sqrt{3}$
    (D) 72
    (E) It cannot be determined from the information given.

12. What is the area of $\triangle ABC$?

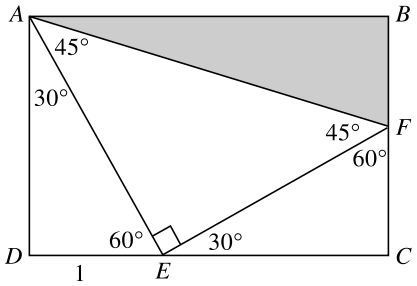
    (A) 108
    (B) $54 + 72\sqrt{2}$
    (C) $54 + 72\sqrt{3}$
    (D) 198
    (E) It cannot be determined from the information given.



13. Which of the following expresses a true relationship between x and y in the figure above?

    (A) $y = 60 - x$ (B) $y = x$ (C) $x + y = 90$
    (D) $y = 180 - 3x$ (E) $x = 90 - 3y$

Questions 14 and 15 refer to the following figure, in which rectangle *ABCD* is divided into two 30-60-90 triangles, a 45-45-90 triangle, and shaded triangle *ABF*.



14. What is the perimeter of shaded triangle *ABF*?

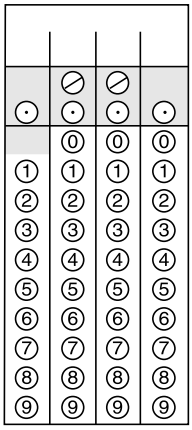
    (A) $\sqrt{2} + 2\sqrt{3}$ (B) $1 + \sqrt{2} + \sqrt{3}$
    (C) $2 + \sqrt{2} + \sqrt{3}$ (D) $2 + 2\sqrt{2} + \sqrt{3}$
    (E) $2\sqrt{2} + 2\sqrt{3}$

15. What is the area of shaded triangle *ABF*?

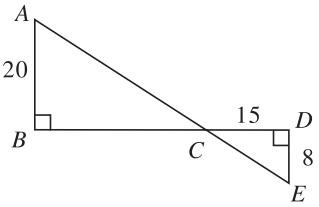
    (A) $\frac{\sqrt{3}}{2}$ (B) 1 (C) $\frac{2\sqrt{3}}{3}$ (D) $\frac{\sqrt{3}+1}{2}$
    (E) $\sqrt{2}\left(\sqrt{3}+1\right)$

Grid-in Questions

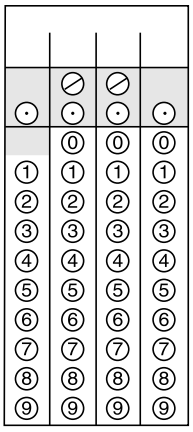
16. If the difference between the measures of the two smaller angles of a right triangle is 20°, what is the measure, in degrees, of the smallest angle?



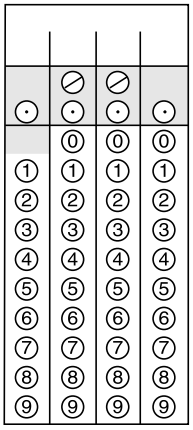
Questions 17 and 18 refer to the figure below.



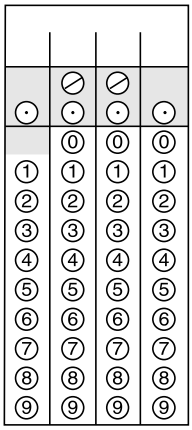
17. What is the perimeter of $\triangle ABC$?



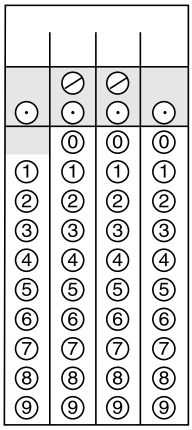
18. What is the area of $\triangle ABC$?



19. What is the smallest integer, x, for which x, $x + 5$, and $2x - 15$ can be the lengths of the sides of a triangle?

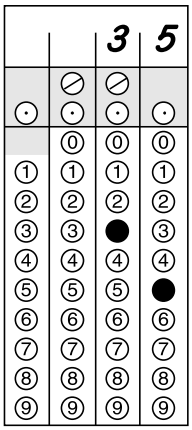


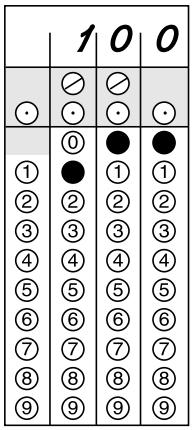
20. If the measures of the angles of a triangle are in the ratio of 1:2:3, and if the perimeter of the triangle is $30 + 10\sqrt{3}$, what is the length of the smallest side?

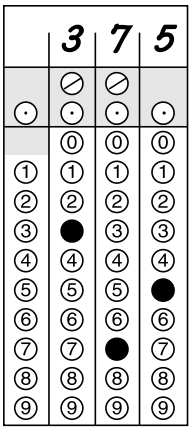


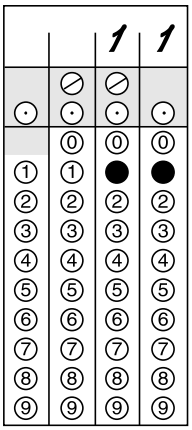
Answer Key

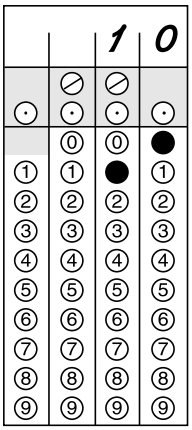
1.	**D**	4.	**D**	7.	**B**	10.	**B**	13.	**A**
2.	**B**	5.	**C**	8.	**D**	11.	**C**	14.	**E**
3.	**D**	6.	**B**	9.	**B**	12.	**C**	15.	**B**

16. 

17. 

18. 

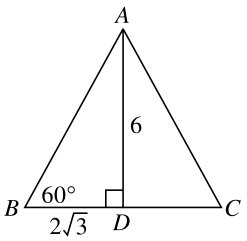
19. 

20. 

Answer Explanations

1. D. $x + 2x + 30 = 180 \Rightarrow 3x + 30 = 180 \Rightarrow 3x = 150 \Rightarrow x = 50$.

2. B. Draw altitude $\overline{AD}$ in equilateral triangle ABC.



By KEY FACT J11:

$$BD = \frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3},$$

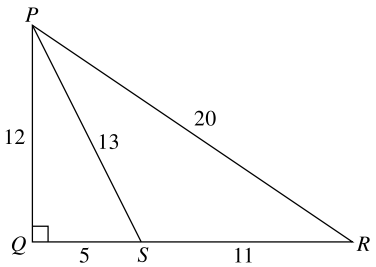
and BD is one-half the base. The area is $2\sqrt{3} \times 6 = 12\sqrt{3}$.

3. D. If the triangle were not required to be right, by KEY FACTS J11 and J12 *any* number greater than 1 and less than 25 could be the length of the third side. For a right triangle, however, there are only *two* possibilities.
(i) If 13 is the hypotenuse, then the legs are 12 and 5. (I is true.) (If you didn't recognize a 5-12-13 triangle, use the Pythagorean theorem: $5^2 + x^2 = 13^2$, and solve.)
(ii) If 12 and 13 are the two legs, then use the Pythagorean theorem to find the length of the hypotenuse:

$$12^2 + 13^2 = c^2 \Rightarrow c^2 = 144 + 169 = 313 \Rightarrow c = \sqrt{313}.$$

(III is true.)
An 11-12-13 triangle is not a *right* triangle. (II is false.)

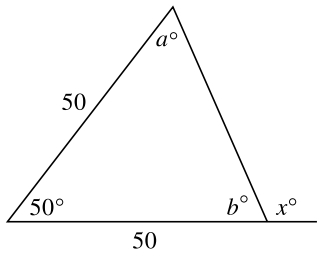
4. D. Use the Pythagorean theorem twice, unless you recognize the common right triangles in this figure (*which you should*). Since $PR = 20$ and $QR = 16$, $\triangle PQR$ is a $3x$-$4x$-$5x$ right triangle with $x = 4$. Then $PQ = 12$, and $\triangle PQS$ is a right triangle whose legs are 5 and 12. The hypotenuse, PS, therefore, is 13. [If you had difficulty with this question, review the material, but in the meantime remember TACTIC 2: trust the diagram. $\overline{PS}$ is longer than $\overline{SR}$, so you can eliminate A, B, and C, and $\overline{PS}$ is clearly shorter than $\overline{QR}$, so eliminate E.]



5. C. Here,

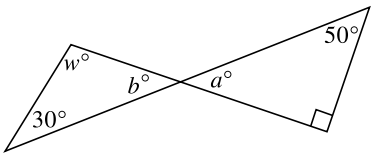
$$50 + a + b = 180 \Rightarrow a + b = 130,$$

and since the triangle is isosceles, $a = b$. Therefore, a and b are each 65, and $x = 180 - 65 = 115$.



6. B. Here, $50 + 90 + a = 180 \Rightarrow a = 40$, and since vertical angles are equal, $b = 40$. Then:

$$40 + 30 + w = 180 \Rightarrow w = 110.$$



7. B. You *could* calculate the area of the rectangle and subtract the areas of the two white right triangles, but don't. The shaded area is a triangle whose base is 4 and whose height is 12. The area is $\frac{1}{2}(4)(12) = 24$.

8. D. Since both $\overline{BD}$ and $\overline{ED}$ are the hypotenuses of right triangles, their lengths can be calculated by the Pythagorean theorem, but again these are triangles you should recognize: the sides of $\triangle DCE$ are 5-12-13, and those of $\triangle BAD$ are 9-12-15 ($3x$-$4x$-$5x$, with $x = 3$). Therefore, the perimeter of $\triangle BED$ is $4 + 13 + 15 = 32$.

9. B. Since $\triangle DGH$ is a right triangle, whose hypotenuse is 5 and one of whose legs is 3, the other leg, GH, is 4. Since $GF = DE = 7$, $HF = 3$. Now, $\triangle DFH$ has a base of 3 (HF) and a height of 3 (DG), and its area is $\frac{1}{2}(3)(3) = 4.5$.

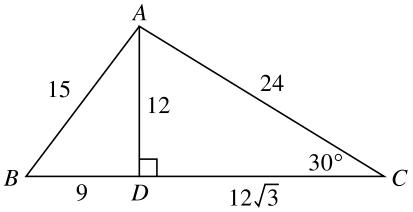
10. B. For $\triangle DFH$, you already have that $DH = 5$ and $HF = 3$; you need only find DF, which is the hypotenuse of $\triangle DEF$. By the Pythagorean theorem,

$$3^2 + 7^2 = (DF)^2 \Rightarrow (DF)^2 = 9 + 49 = 58 \Rightarrow DF = \sqrt{58}.$$

The perimeter is $3 + 5 + \sqrt{58} = 8 + \sqrt{58}$.

11. C. Triangle ADB is a right triangle whose hypotenuse is 15 and one of whose legs is 9, so this is a $3x$-$4x$-$5x$ triangle with $x = 3$, and $AD = 12$. Now $\triangle ADC$ is a 30-60-90 triangle, whose shorter leg is 12. Hypotenuse AC is 24, and leg CD is $12\sqrt{3}$, so the perimeter is

$$24 + 15 + 9 + 12\sqrt{3} = 48 + 12\sqrt{3}.$$

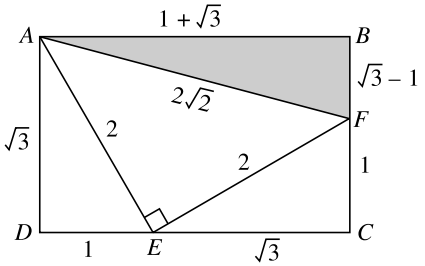


12. C. From the solution to Exercise 11, you have the base $(9 + 12\sqrt{3})$ and the height (12) of $\triangle ABC$. Then, the area is

$$\frac{1}{\cancel{2}_{1}}\cancel{(12)}^{6}(9 + 12\sqrt{3}) = 54 + 72\sqrt{3}.$$

13. A. $x + 2x + 3y = 180 \Rightarrow 3x + 3y = 180 \Rightarrow$ $x + y = 60 \Rightarrow y = 60 - x$.

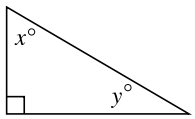
14. E.



You are given enough information to determine the sides of all the triangles. Both 30-60-90 triangles have sides 1, $\sqrt{3}$, 2; and the 45-45-90 triangle has sides 2, 2, $2\sqrt{2}$. Also, $AB = CD = 1 + \sqrt{3}$, and $BF = AD - CF = \sqrt{3} - 1$. Then, the perimeter of the shaded triangle is $1 + \sqrt{3} + \sqrt{3} - 1 + 2\sqrt{2} = 2\sqrt{2} + 2\sqrt{3}$.

15. B. The area of $\Delta ABF = \frac{1}{2}(\sqrt{3} + 1)(\sqrt{3} - 1) =$ $\frac{1}{2}(3 - 1) = \frac{1}{2}(2) = 1$.

16. (35)



- Draw a diagram and label it. Write the equations, letting x = larger angle and y = smaller angle:
$$\begin{array}{rr} x + y = & 90 \\ \underline{+\ x - y = } & \underline{20} \\ 2x & = 110 \end{array}$$
- Add the equations:

Then $x = 55$, and $y = 90 - 55 = 35$.

17. (100) By the Pythagorean theorem, $8^2 + 15^2 = (CE)^2 \Rightarrow$ $(CE)^2 = 64 + 225 = 289$ $\Rightarrow CE = 17$.

Then the perimeter of $\triangle CDE = 8 + 15 + 17 = 40$. Triangles ABC and CDE are similar (each has a 90° angle, and the vertical angles at C are congruent). The ratio of similitude is $\frac{20}{8} = 2.5$, so the perimeter of $\triangle ABC$ is $2.5 \times 40 = 100$.

18. (375) The area of $\triangle CDE = \frac{1}{2}(8)(15) = 60$.

Since the ratio of similitude for the two triangles (as calculated in Solution 17) is 2.5, the area of $\triangle ABC$ is $(2.5)^2$ times the area of $\triangle CDE$: $(2.5)^2 \times 60 = 6.25 \times 60 = 375$.

19. (11) In a triangle the sum of the lengths of any two sides must be greater than the third side. For $x + (x + 5)$ to be greater than $2x - 15$, $2x + 5$ must be greater than $2x - 15$; but that's always true. For $x + (2x - 15)$ to be greater than $x + 5$, $3x - 15$ must be greater than $x + 5$; but $3x - 15 > x + 5$ is true only if $2x > 20$, which means $x > 10$. Grid in 11.

20. (10) If the measures of the angles are in the ratio of 1:2:3, then:

$$x + 2x + 3x = 180 \Rightarrow 6x = 180 \Rightarrow x = 30.$$

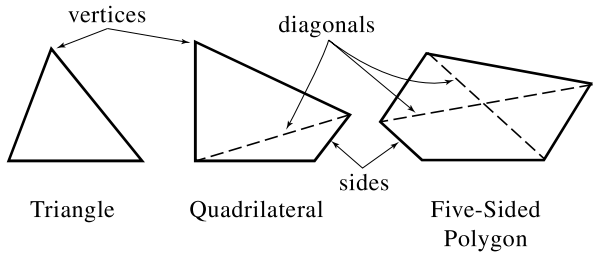
The triangle is a 30-60-90 right triangle, and the sides are a, $2a$, and $a\sqrt{3}$. The perimeter therefore is $3a + a\sqrt{3} = a(3 + \sqrt{3})$, so

$$a(3 + \sqrt{3}) = 30 + 10\sqrt{3} = 10(3 + \sqrt{3}) \Rightarrow$$
$$a = 10.$$

12-K QUADRILATERALS AND OTHER POLYGONS

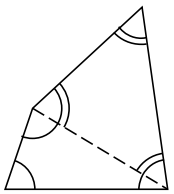
A ***polygon*** is a closed geometric figure made up of line segments. The line segments are called ***sides***, and the endpoints of the line segments are called ***vertices*** (each one is a ***vertex***). Line segments inside the polygon drawn from one vertex to another are called ***diagonals***.

The simplest polygons, which have three sides, are the triangles, which you studied in Section 12-J. A polygon with four sides is called a ***quadrilateral***. There are special names (such as *pentagon* and *hexagon*) for polygons with more than four sides, but you do not need to know any of them for the SAT.



This section will present a few facts about polygons in general and then review the key facts you need to know about three special quadrilaterals.

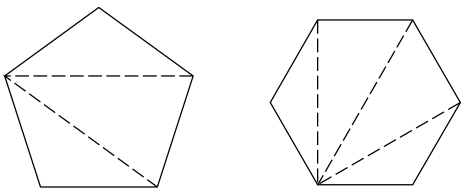
Every quadrilateral has two diagonals. If you draw in either one, you will divide the quadrilateral into two triangles. Since the sum of the measures of the three angles in each of the triangles is 180°, the sum of the measures of the angles in the quadrilateral is 360°.



Key Fact K1

In any quadrilateral, the sum of the measures of the four angles is 360°.

In exactly the same way as shown above, any polygon can be divided into triangles by drawing in all of the diagonals emanating from one vertex.



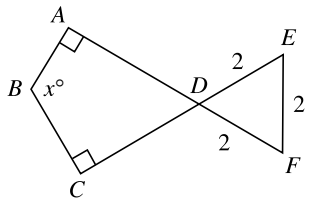
Notice that a five-sided polygon is divided into three triangles, and a six-sided polygon is divided into four triangles. In general, an *n*-sided polygon is divided into $(n - 2)$ triangles, which leads to KEY FACT K2.

Key Fact K2

The sum of the measures of the n angles in a polygon with n sides is $(n - 2) \times 180°$.

Example 1.

In the figure below, what is the value of x?

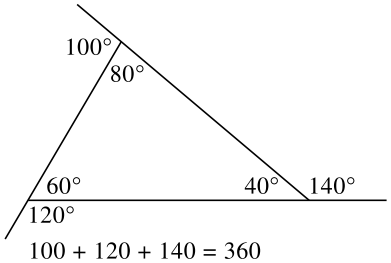


Solution. Since ΔDEF is equilateral, all of its angles measure 60°; also, since the two angles at vertex D are vertical angles, their measures are equal. Therefore, the measure of $\angle D$ in quadrilateral *ABCD* is 60°. Also, $\angle A$ and $\angle C$ are right angles, so each measures 90°.

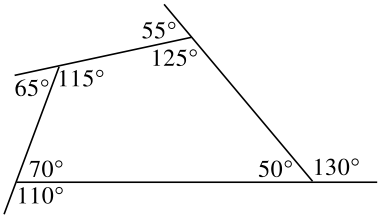
Finally, since the sum of the measures of all four angles of *ABCD* is 360°:

$$60 + 90 + 90 + x = 360 \Rightarrow 240 + x = 360 \Rightarrow x = \mathbf{120}.$$

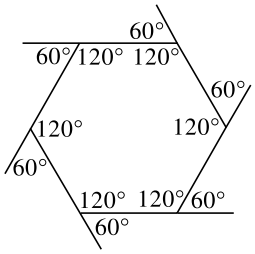
An ***exterior angle*** of a polygon is formed by extending a side. In the polygons below, one exterior angle has been drawn in at each vertex. Surprisingly, if you add the measures of all of the exterior angles in any of the polygons, the sums are equal.



100 + 120 + 140 = 360



65 + 110 + 130 + 55 = 360



60 + 60 + 60 + 60 + 60 + 60 = 360

Key Fact K3

In any polygon, the sum of the measures of the exterior angles, taking one at each vertex, is 360°.

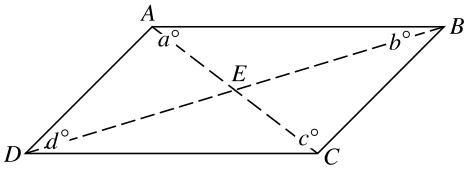
Example 2.

A 10-sided polygon is drawn in which each angle has the same measure. What is the measure, in degrees, of each angle?

Solution 1. By KEY FACT K2, the sum of the degree measures of the 10 angles is $(10 - 2) \times 180 = 8 \times 180 = 1440$. Then, each angle is $1440 \div 10 =$ **144**.

Solution 2. By KEY FACT K3, the sum of the 10 exterior angles is 360, so each one is 36. Therefore, each interior angle is $180 - 36 =$ **144**.

A ***parallelogram*** is a quadrilateral in which both pairs of opposite sides are parallel.



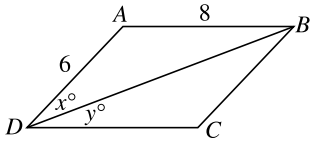
Key Fact K4

Parallelograms have the following properties:

- **Opposite sides are congruent: $AB = CD$ and $AD = BC$.**
- **Opposite angles are congruent: $a = c$ and $b = d$.**
- **Consecutive angles add up to 180°:**
  $a + b = 180$, $b + c = 180$, $c + d = 180$, and a $+ d = 180$.
- **The two diagonals bisect each other:**
  $AE = EC$ and $BE = ED$.
- **A diagonal divides the parallelogram into two triangles that have exactly the same size and shape. (The triangles are congruent.)**

Example 3.

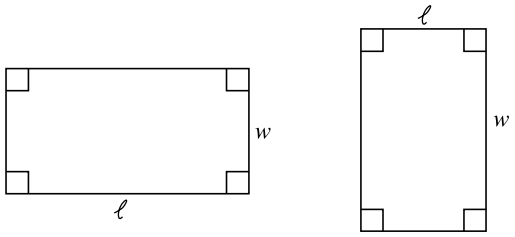
In the figure below, $ABCD$ is a parallelogram. Which of the following statements *must* be true?



(A) $x < y$ (B) $x = y$ (C) $x > y$ (D) $x + y < 90$
(E) $x + y > 90$

Solution. Since $\overline{AB}$ and $\overline{CD}$ are parallel line segments cut by transversal $\overline{BD}$, $m\angle ABD = y$. In $\triangle ABD$, $AB > AD$, so by KEY FACT J3 the measure of the angle opposite $\overline{AB}$ is greater than the measure of the angle opposite $\overline{AD}$. Therefore, ***x > y* (C)**.

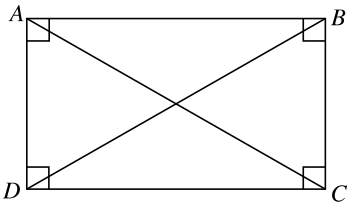
A ***rectangle*** is a parallelogram in which all four angles are right angles. Two adjacent sides of a rectangle are usually called the ***length*** (ℓ) and the ***width*** (w). Note that the length is not necessarily greater than the width.



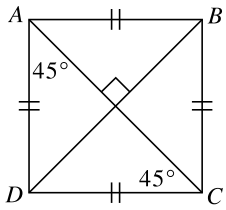
Key Fact K5

Since a rectangle is a parallelogram, all of the properties listed in KEY FACT K4 hold for rectangles. In addition:

- **The measure of each angle in a rectangle is 90°.**
- **The diagonals of a rectangle are congruent: $AC \cong BD$.**



A ***square*** is a rectangle in which all four sides have the same length.



Key Fact K6

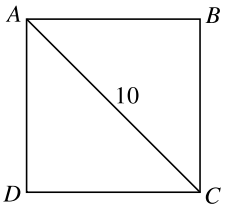
Since a square is a rectangle, all of the properties listed in KEY FACTS K4 and K5 hold for squares. In addition:

- **All four sides have the same length.**
- **Each diagonal divides the square into two 45-45-90 right triangles.**
- **The diagonals are perpendicular to each other:**
  $\overline{AC} \perp \overline{BD}$.

Example 4.

What is the length of each side of a square if its diagonals are 10?

Solution. Draw a diagram. In square *ABCD*, diagonal $\overline{AC}$ is the hypotenuse of a 45-45-90 right triangle, and side $\overline{AB}$ is a leg of that triangle. By KEY FACT J7,

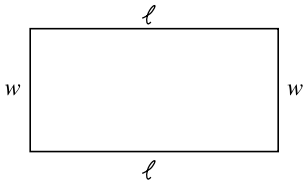


$$AB = \frac{AC}{\sqrt{2}} = \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{10\sqrt{2}}{2} = \mathbf{5\sqrt{2}.}$$

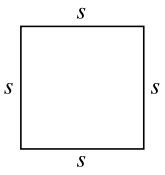
The ***perimeter*** (*P*) of any polygon is the sum of the lengths of all of its sides. The only polygons for which we have formulas for the perimeter are the rectangle and the square.

Key Fact K7

In a rectangle, $P = 2(\ell + w)$; in a square, $P = 4s$.



$P = \ell + w + \ell + w = 2(\ell + w)$

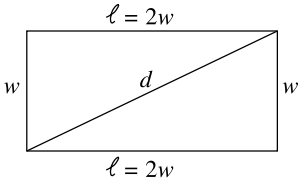


$P = s + s + s + s = 4s$

Example 5.

The length of a rectangle is twice its width. If the perimeter of the rectangle is the same as the perimeter of a square of side 6, what is the square of the length of a diagonal of the rectangle?

Solution. Don't do anything until you have drawn diagrams. Since the perimeter of the square is 24, the perimeter of the rectangle is also 24. Then

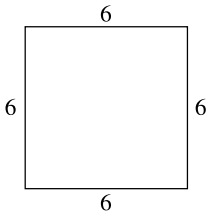


$$2(\ell + w) = 24 \Rightarrow$$
$$\ell + w = 12.$$

But $\ell = 2w$, so

$$2w + w = 3w = 12 \Rightarrow$$
$$w = 4 \text{ (and } \ell = 8).$$

Finally, use the Pythagorean theorem:
$d^2 = 4^2 + 8^2 = 16 + 64 = \mathbf{80}$.



$P = 24$

In Section 12-J you reviewed the formula for the *area* of a triangle. The only other polygons for which you need to know area formulas are the parallelogram, rectangle, and square.

- Parallelogram: Since the area of each of the two triangles formed by drawing a diagonal in a parallelogram is $\frac{1}{2}bh$, the area of the parallelogram is twice as great:

$$A = \frac{1}{2}\ bh + \frac{1}{2}\ bh = bh.$$

- Rectangle: In a rectangle the same formula holds, but it is usually written as

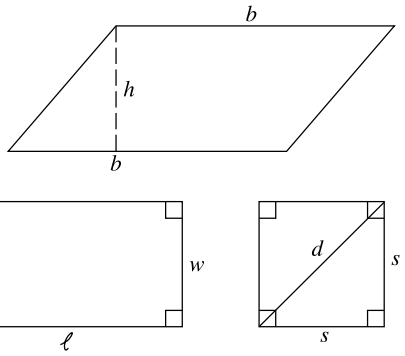
$$A = \ell w,$$

using the terms *length* and *width* instead of *base* and *height*.
- Square: In a square the length and width are equal; we label each of them *s* (side), and write

$$A = s \times s = s^2.$$

If *d* is the diagonal of a square,

$$d = s\sqrt{2} \ \Rightarrow d^2 = 2s^2 \Rightarrow s^2 = \frac{1}{2}\ d^2.$$



Key Fact K8

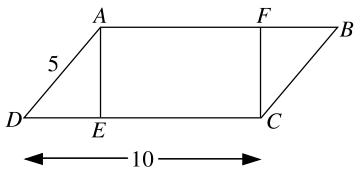
Here are the area formulas you need to know:

- **For a parallelogram: $A = bh$.**
- **For a rectangle: $A = \ell w$.**
- **For a square: $A = s^2$ or $A = \frac{1}{2}d^2$.**

The formula for the area of a rectangle is one of the facts provided in the "Reference Information" at the beginning of each math section.

Example 6.

In the figure below, the area of parallelogram *ABCD* is 40. What is the area of rectangle *AFCE*?

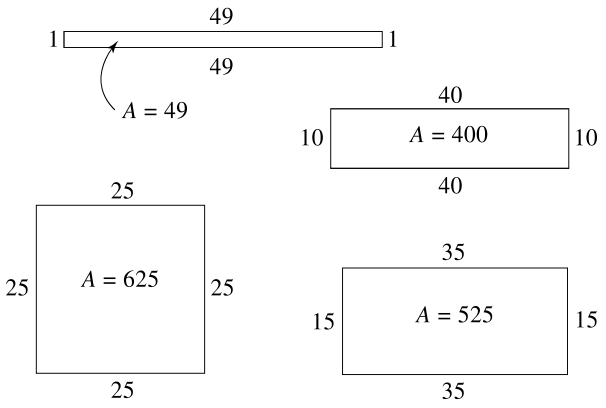


(A) 20 (B) 24 (C) 28 (D) 32 (E) 36

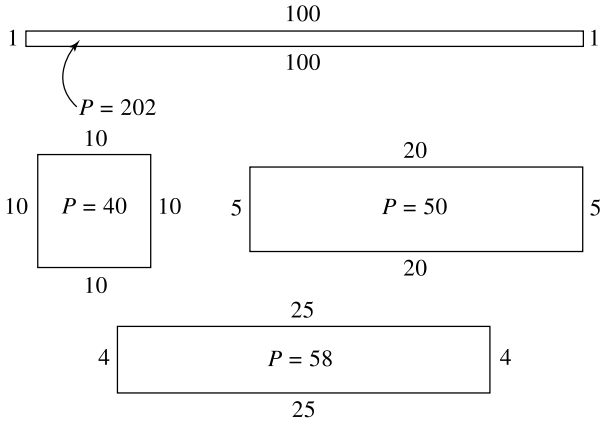
Solution. Since the base, *CD*, is 10 and the area is 40, the height, *AE*, must be 4. Then $\triangle AED$ must be a 3-4-5 right triangle with $DE = 3$, which implies that $EC = 7$. The area of the rectangle is $7 \times 4 = \mathbf{28\ (C).}$

Two rectangles with the same perimeter can have different areas, and two rectangles with the same area can have different perimeters. These facts are a common source of questions on the SAT.

RECTANGLES WHOSE PERIMETERS ARE 100



RECTANGLES WHOSE AREAS ARE 100



Key Fact K9

For a given perimeter, the rectangle with the largest area is a square. For a given area, the rectangle with the smallest perimeter is a square.

Example 7.

The area of rectangle I is 10, and the area of rectangle II is 12. Which of the following statements could be true?

I. Perimeter of rectangle I < perimeter of rectangle II.

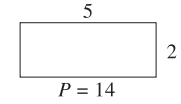
II. Perimeter of rectangle I = perimeter of rectangle II.

III. Perimeter of rectangle I > perimeter of rectangle II.

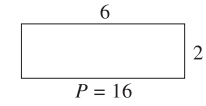
(A) I only (B) II only (C) I and II only
(D) I and III only (E) I, II, and III

Solution.

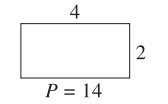
- If the dimensions of rectangle I are 5×2 and the dimensions of rectangle II are 6×2, then the perimeters are 14 and 16, respectively. (I is true.)

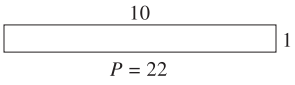


- If rectangle I is 5×2 and rectangle II is 3×4, then both perimeters are 14. (II is true.)



- If the dimensions of rectangle I are 10×1, its perimeter is 22, which is greater than the perimeter of rectangle II, above. (III is true.)



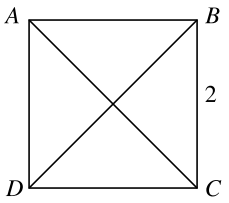


Statements **I, II, and III** are true **(E)**.

Exercises on Quadrilaterals and Other Polygons

Multiple-Choice Questions

1. In the figure at the right, the two diagonals divide square *ABCD* into four small triangles. What is the sum of the perimeters of those triangles?



(A) $2 + 2\sqrt{2}$ (B) $8 + 4\sqrt{2}$
(C) $8 + 8\sqrt{2}$ (D) 16
(E) 24

2. If the length of a rectangle is 4 times its width, and if its area is 144, what is its perimeter?

(A) 6 (B) 24 (C) 30 (D) 60 (E) 96

3. If the angles of a five-sided polygon are in the ratio of 2:3:3:5:5, what is the degree measure of the smallest angle?

(A) 20 (B) 40 (C) 60 (D) 80 (E) 90

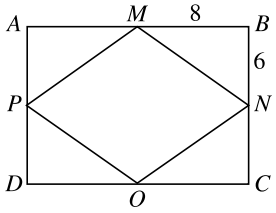
Questions 4 and 5 refer to a rectangle in which the length of each diagonal is 12, and one of the angles formed by the diagonal and a side measures 30°.

4. What is the area of the rectangle?
   (A) 18 (B) 72 (C) $18\sqrt{3}$ (D) $36\sqrt{3}$ (E) $36\sqrt{2}$

5. What is the perimeter of the rectangle?
   (A) 18 (B) 24 (C) $12 + 12\sqrt{3}$ (D) $18 + 6\sqrt{3}$ (E) $24\sqrt{2}$

6. The length of a rectangle is 5 more than the side of a square, and the width of the rectangle is 5 less than the side of the square. If the area of the square is 45, what is the area of the rectangle?
   (A) 20 (B) 25 (C) 45 (D) 50 (E) 70

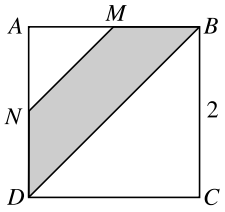
Questions 7 and 8 refer to the following figure, in which M, N, O, and P are the midpoints of the sides of rectangle $ABCD$.



7. What is the perimeter of quadrilateral $MNOP$?
   (A) 24 (B) 32 (C) 40 (D) 48 (E) 60

8. What is the area of quadrilateral $MNOP$?
   (A) 48 (B) 60 (C) 72 (D) 96 (E) 108

Questions 9 and 10 refer to the following figure, in which M and N are midpoints of two of the sides of square $ABCD$.

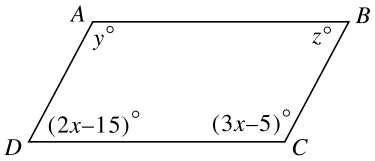


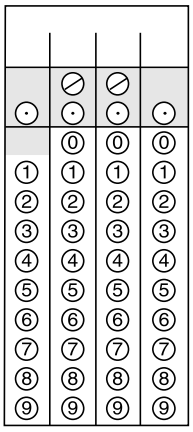
9. What is the perimeter of the shaded region?
   (A) 3 (B) $2 + 3\sqrt{2}$ (C) $3 + 2\sqrt{2}$ (D) 5 (E) 8

10. What is the area of the shaded region?
    (A) 1.5 (B) 1.75 (C) 3 (D) $2\sqrt{2}$ (E) $3\sqrt{2}$

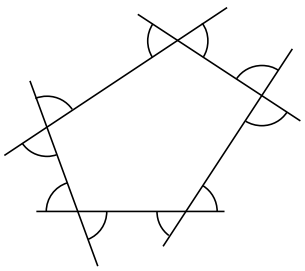
Grid-in Questions

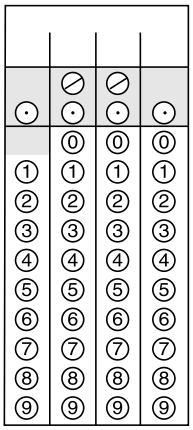
11. In the figure below, $ABCD$ is a parallelogram. What is the value of $y - z$?



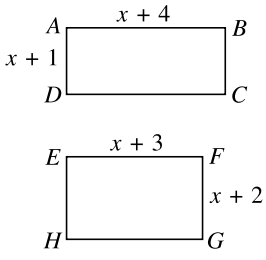


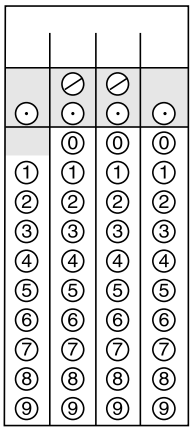
12. In the figure below, what is the sum of the degree measures of all of the marked angles?



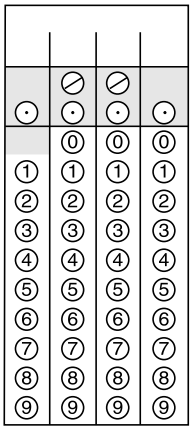


13. If, in the figures below, the area of rectangle $ABCD$ is 100, what is the area of rectangle $EFGH$?

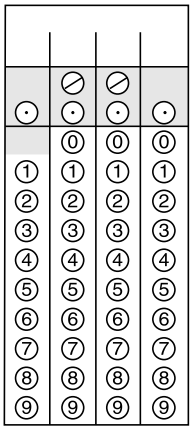




14. How many sides does a polygon have if the measure of each interior angle is 8 times the degree measure of each exterior angle?

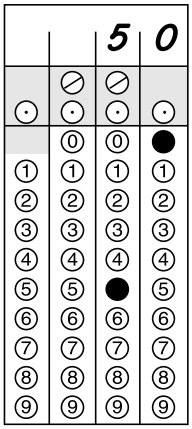


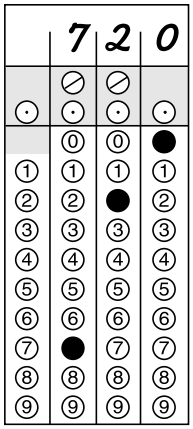
25. In quadrilateral $WXYZ$, the measure of $\angle Z$ is 10 more than twice the average of the measures of the other three angles. What is the measure, in degrees, of $\angle Z$?

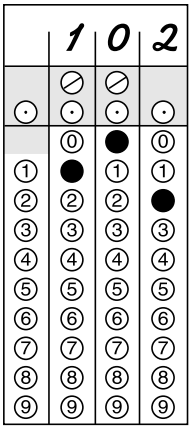


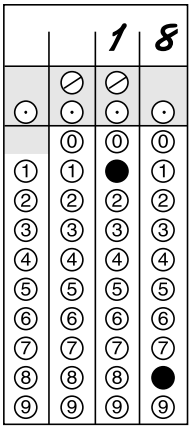
Answer Key

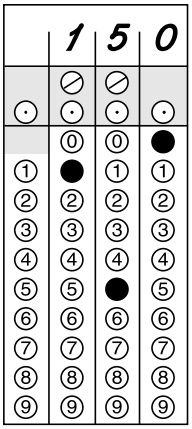
1.	**C**	3.	**C**	5.	**C**	7.	**C**	9.	**B**
2.	**D**	4.	**D**	6.	**A**	8.	**D**	10.	**A**

11. 

12. 

13. 

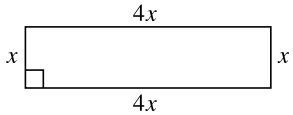
14. 

15. 

Answer Explanations

1. C. Each of the four small triangles is a 45-45-90 right triangle whose hypotenuse is 2. Therefore, each leg is $\frac{2}{\sqrt{2}} = \sqrt{2}$. The perimeter of each small triangle is $2 + 2\sqrt{2}$, and the sum of the perimeters is 4 times as great: $8 + 8\sqrt{2}$.

2. D. Draw a diagram and label it.



Since the area is 144, then

$$144 = 4x^2 \Rightarrow x^2 = 36 \Rightarrow x = 6.$$

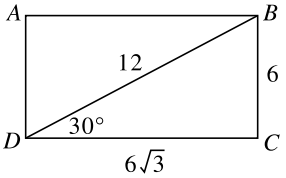
The width is 6, the length is 24, and the perimeter is 60.

3. C. The sum of the degree measures of the angles of a five-sided polygon is $(5 - 2) \times 180 = 3 \times 180 = 540$. Then:

$$540 = 2x + 3x + 3x + 5x + 5x = 18x \Rightarrow$$
$$x = 540 \div 18 = 30.$$

The degree measure of the smallest angle is $2x$: $2 \times 30 = 60$.

4. D. Draw a diagram and label it. Since $\triangle BCD$ is a 30-60-90 right triangle, BC is 6 (half the hypotenuse) and CD is $6\sqrt{3}$. Then the area of rectangle $ABCD$ is $\ell w = 6(6\sqrt{3}) = 36\sqrt{3}$.



5. C. The perimeter of the rectangle is $2(\ell + w) = 2(6 + 6\sqrt{3}) = 12 + 12\sqrt{3}$.

6. A. Let x represent the side of the square. Then the dimensions of the rectangle are $(x + 5)$ and $(x - 5)$, and its area is $(x + 5)(x - 5) = x^2 - 25$. Since 45 is the area of the square, $x^2 = 45$, and so $x^2 - 25 = 20$.

7. C. Each triangle surrounding quadrilateral *MNOP* is a 6-8-10 right triangle. Then, each side of the quadrilateral is 10, and its perimeter is 40.

8. D. The area of each of the triangles is $\frac{1}{2}(6)(8) = 24$, so together the four triangles have an area of 96. The area of the rectangle is $16 \times 12 = 192$. Therefore, the area of quadrilateral *MNOP* is $192 - 96 = 96$.
NOTE: Joining the midpoints of the four sides of any quadrilateral creates a parallelogram whose area is one-half the area of the original quadrilateral.

9. B. Since *M* and *N* are midpoints of sides of length 2, *AM*, *MB*, *AN*, and *ND* are each equal to 1. Also, $MN = \sqrt{2}$, since it's the hypotenuse of an isosceles right triangle whose legs are 1; and $BD = 2\sqrt{2}$, since it's the hypotenuse of an isosceles right triangle whose legs are 2. Then, the perimeter of the shaded region is $1 + \sqrt{2} + 1 + 2\sqrt{2} = 2 + 3\sqrt{2}$.

10. A. The area of $\triangle ABD = \frac{1}{2}(2)(2) = 2$, and the area of $\triangle AMN = \frac{1}{2}(1)(1) = 0.5$. The area of the shaded region is $2 - 0.5 = 1.5$.

11. (50) The sum of the degree measures of two consecutive angles of a parallelogram is 180, so

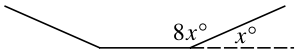
$$180 = (3x - 5) + (2x - 15) = 5x - 20 \Rightarrow 5x = 200 \Rightarrow x = 40.$$

Since opposite angles of a parallelogram are equal, $y = 3x - 5 = 115$ and $z = 2x - 15 = 65$. Then $y - z = 50$.

12. (720) Each of the 10 marked angles is an exterior angle of the pentagon. If you take one angle at each vertex, the sum of the degree measures of those five angles is 360; the sum of the degree measures of the other five is also 360: $360 + 360 = 720$.

13. (102) The area of rectangle *ABCD* = $(x + 1)(x + 4) = x^2 + 5x + 4$. The area of rectangle *EFGH* = $(x + 2)(x + 3) = x^2 + 5x + 6$, which is exactly 2 more than the area of rectangle *ABCD*: $100 + 2 = 102$.

14. (18) The sum of the degree measures of an interior and exterior angle is 180, so $180 = 8x + x = 9x \Rightarrow x = 20$. Since the sum of the degree measures of all the exterior angles of a polygon is 360, there are $360 \div 20 = 18$ angles and, of course, 18 sides.



15. (150) Let *W*, *X*, *Y*, and *Z* represent the degree measures of the four angles. Since $W + X + Y + Z = 360$, then $W + X + Y = 360 - Z$. Also:

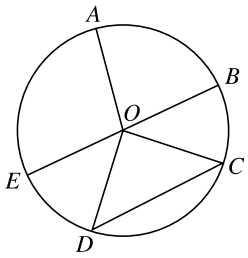
$$Z = 10 + 2\left(\frac{W + X + Y}{3}\right) = 10 + 2\left(\frac{360 - Z}{3}\right).$$

Then:

$$Z = 10 + \frac{2}{3}(360) - \frac{2}{3}Z = 10 + 240 - \frac{2}{3}Z$$
$$\Rightarrow \frac{5}{3}Z = 250 \Rightarrow Z = 150.$$

12-L CIRCLES

A ***circle*** consists of all the points that are the same distance from one fixed point, called the ***center***. That distance is called the ***radius*** of the circle. The figure at the right is a circle of radius 1 unit whose center is at point *O*. *A*, *B*, *C*, *D*, and *E*, which are each 1 unit from *O*, are all points on circle *O*. The word ***radius*** is also used to represent any of the line segments joining the center and a point on the circle. The plural of *radius* is ***radii***. In circle *O*, above, $\overline{OA}$, $\overline{OB}$, $\overline{OC}$, $\overline{OD}$, and $\overline{OE}$ are all radii. If a circle has radius *r*, each of the radii is *r* units long.

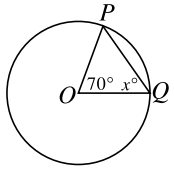


Key Fact L1

Any triangle, such as $\triangle COD$ in the figure above, formed by connecting the endpoints of two radii is isosceles.

Example 1.

If *P* and *Q* are points on circle *O*, what is the value of *x*?



Solution. Since $\overline{OP} \cong \overline{OQ}$ (each is a radius of the circle), $\triangle POQ$ is isosceles. Then $\angle P$ and $\angle Q$ are congruent, so

$$70 + x + x = 180 \Rightarrow 2x = 110 \Rightarrow x = \mathbf{55}.$$

A line segment, such as $\overline{BE}$ in circle *O* at the beginning of this section, whose endpoints are on a circle and that passes through the center is called a ***diameter***. Since $\overline{BE}$ is made up of two radii, $\overline{OB}$ and $\overline{OE}$, a diameter is twice as long as a radius.

Key Fact L2

If d is the diameter and r the radius of a circle, then $d = 2r$.

Key Fact L3

A diameter is the longest line segment that can be drawn in a circle.

The total length around a circle, from A to B to C to D to E and back to A in the circle at the beginning of this section, is called the ***circumference*** of the circle. In every circle the ratio of the circumference to the diameter is exactly the same and is denoted by the symbol π (the Greek letter pi).

Key Fact L4

For every circle:

$$\pi = \frac{\text{circumference}}{\text{diameter}} = \frac{C}{d} \quad \text{or} \quad C = \pi d \quad \text{or} \quad C = 2\pi r.$$

The formula for the circumference of a circle is one of the facts provided in the "Reference Information" at the beginning of each math section.

Key Fact L5

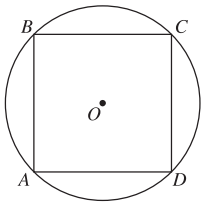
The value of π is *approximately* 3.14.

CALCULATOR HINT

On almost every question on the SAT that involves circles, you are expected to leave your answer in terms of π, so don't multiply by 3.14 unless you must. If you need an approximation—to test a choice, for example—then use your calculator. If you have a scientific calculator, use the π key. This not only is faster and more accurate than punching in 3.14, but also avoids careless mistakes in entering.

Example 2.

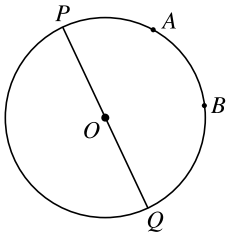
In the figure at the right, square $ABCD$ is inscribed in circle O. If the area of the square is 50, what is the circumference of the circle?



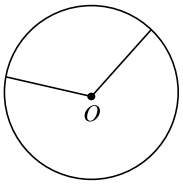
(A) $\pi\sqrt{50}$ (B) 10π (C) 25π (D) 50π (E) 100π

Solution. Since the area of square $ABCD$ is 50, the length of each side is $\sqrt{50}$. Diagonal $\overline{AC}$ divides the square into two isosceles right triangles whose legs are $\sqrt{50}$ and whose hypotenuse is $\overline{AC}$. So, $\overline{AC} = \left(\sqrt{50}\right)\left(\sqrt{2}\right) = \sqrt{100} = 10$. But since $\overline{AC}$ is also a diameter of circle O, the circumference is $\pi d = \mathbf{10\pi}$ **(B)**.

An ***arc*** consists of two points on a circle and all the points between them. If two points, such as P and Q in circle O, are the endpoints of a diameter, they divide the circle into two arcs called ***semicircles***. On the SAT, *arc AB* always refers to the small arc joining A and B. To refer to the large arc going from A to B through P and Q, we would say *arc APB* or *arc AQB*.

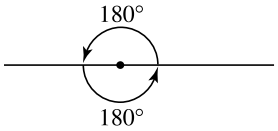


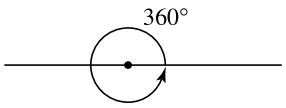
An angle whose vertex is at the center of a circle is called a ***central angle***.



Key Fact L6

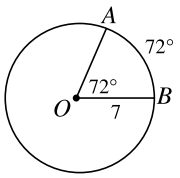
The degree measure of a complete circle is 360.

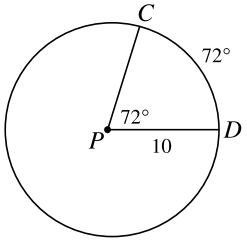




Key Fact L7

The degree measure of an arc equals the degree measure of the central angle that intercepts it.





CAUTION: Degree measure is *not* a measure of length. In the circles above, arc AB and arc CD each measure 72°, even though arc CD is much longer.

How long *is* arc CD? Since the radius of circle P is 10, its circumference is 20π [$2\pi r = 2\pi(10) = 20\pi$]. Since there are 360° in a circle, arc CD is $\frac{72}{360}$, or $\frac{1}{5}$, of the circumference: $\frac{1}{5}(20\pi) = 4\pi$.

Key Fact L8

The formula for the area of a circle of radius r is $A = \pi r^2$.

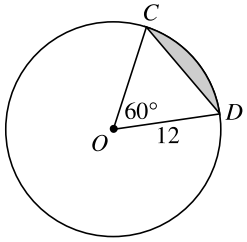
KEY FACT L8 is one of the facts provided in the "Reference Information" at the beginning of each math section.

The area of circle P above is $\pi(10)^2 = 100\pi$ square units. The area of sector CPD is $\frac{1}{5}$ of the area of the circle: $\frac{1}{5}(100\pi) = 20\pi$.

Key Fact L9

If an arc measures $x°$, the length of the arc is $\frac{x}{360}(2\pi r)$; and the area of the sector formed by the arc and two radii is $\frac{x}{360}(\pi r^2)$.

Examples 3 and 4 refer to the circle below.



Example 3.

What is the area of the shaded region?

(A) $144\pi - 144\sqrt{3}$ (B) $144\pi - 36\sqrt{3}$ (C) $144\pi - 72$
(D) $24\pi - 36\sqrt{3}$ (E) $24\pi - 72$

Solution. The area of the shaded region is equal to the area of sector COD minus the area of $\triangle COD$. The area of the circle is $\pi(12)^2 = 144\pi$. Since $\frac{60}{360} = \frac{1}{6}$, the area of sector COD is $\frac{1}{6}(144\pi) = 24\pi$. Since $m\angle O = 60$, $m\angle C + m\angle D = 120$; but $\triangle COD$ is isosceles, so $\angle C = \angle D$. Therefore, each measures 60°, and the triangle is equilateral. Finally, by KEY FACT J15,

$$\text{area of } \triangle COD = \frac{12^2\sqrt{3}}{4} = \frac{144\sqrt{3}}{4} = 36\sqrt{3},$$

so the area of the shaded region is **$24\pi - 36\sqrt{3}$ (D).**

Example 4.

What is the perimeter of the shaded region?

(A) $12 + 4\pi$ (B) $12 + 12\pi$ (C) $12 + 24\pi$
(D) $12\sqrt{2} + 4\pi$ (E) $12\sqrt{2} + 24\pi$

Solution. Since $\triangle COD$ is equilateral, $CD = 12$. Since

$$\text{circumference of circle} = 2\pi(12) = 24\pi \Rightarrow$$
$$\text{arc } CD = \frac{1}{6}(24\pi) = 4\pi,$$

the perimeter is **$12 + 4\pi$ (A).**

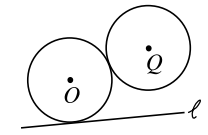
CALCULATOR HINT

If you do Examples 3 and 4 correctly, you won't use your calculator at all (except possibly to divide 144 by 4). On multiple-choice questions involving circles, the choices are almost always in terms of π, so you shouldn't multiply anything by 3.14. However, if you forget how to find the circumference or area of a circle, you will be happy to have a calculator. Suppose that in Example 4 you see that $CD = 12$, but you don't remember how to find the length of arc CD. From the diagram, it is clear that it is slightly longer than CD, say 13, so you know that the perimeter is *about* 25. Now, use your calculator. Evaluate each choice to see which one is closest to 25:

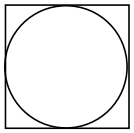
- (A) $12 + 4\pi \approx 12 + 4(3.14) = 24.56$, which is quite close;
- (B) $12 + 12\pi \approx 49.68$, which is way too large;
- (C) is even larger; and (E) is larger yet;
- (D) is 29.5, which isn't as absurd as the other wrong choices, but it is still too large.

The answer is **A**. You could approximate and test the choices in Example 3 in the same way.

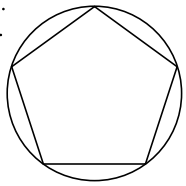
A line and a circle or two circles are ***tangent*** if they have only one point of intersection. A circle is ***inscribed*** in a triangle or square if it is tangent to each side. A polygon is ***inscribed*** in a circle if each vertex is on the circle.



Line ℓ is tangent to circle O.
Circles O and Q are tangent.



The circle is inscribed in the square.



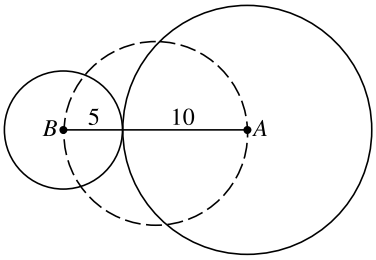
The pentagon is inscribed in the circle.

Example 5.

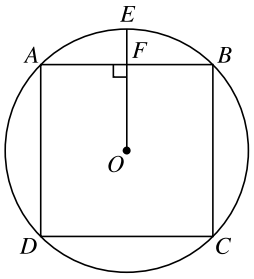
A is the center of a circle whose radius is 10, and B is the center of a circle whose diameter is 10. If these two circles are tangent to one another, what is the area of the circle whose diameter is $\overline{AB}$?

(A) 30π (B) 56.25π (C) 100π (D) 225π (E) 400π

Solution. Draw a diagram. Since the diameter, $\overline{AB}$, of the dotted circle is 15, its radius is 7.5 and its area is $\pi(7.5)^2 =$ **56.25π (B)**. (Note that you should use your calculator to square 7.5 but not to multiply by π.)



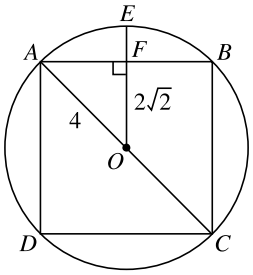
Example 6.



In the figure above, square $ABCD$ is inscribed in a circle whose center is O and whose radius is 4. If $\overline{EO} \perp \overline{AB}$ at F, what is the length of $\overline{EF}$?

(A) 2 (B) $\sqrt{2}$ (C) $2\sqrt{2}$ (D) $4 - \sqrt{2}$ (E) $4 - 2\sqrt{2}$

Solution. Draw diagonal $\overline{AC}$. Then, $\triangle AFO$ is a 45-45-90 right triangle. Since hypotenuse AO is a radius, its length is 4; and by KEY FACT J8:

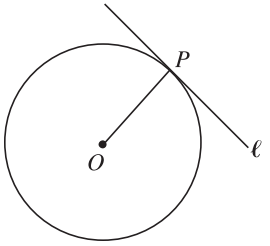


$$OF = \frac{4}{\sqrt{2}} = \frac{4}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = 2\sqrt{2}.$$

$EO = 4$ since it is also a radius. Then

$$EF = EO - OF = \mathbf{4 - 2\sqrt{2}}\ \textbf{(E)}.$$

In the figure below, line ℓ is tangent to circle O at point P.

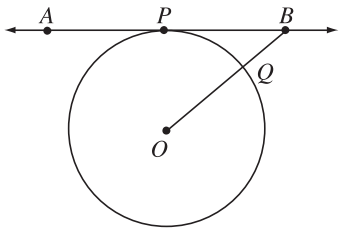


An important theorem in geometry states that radius $\overline{OP}$ is perpendicular to ℓ.

Key Fact L10

A line tangent to a circle is perpendicular to the radius drawn to the point of contact.

Example 7.



In the figure above, $\overleftrightarrow{AB}$ is tangent to circle O at point P. If $OB = 15$ and $PB = 12$, what is QB?

Solution. Draw in radius $\overline{OP}$, creating $\triangle OPB$. By KEY FACT L10, $\overline{OP} \perp \overline{AB}$, so $\triangle OPB$ is a right triangle. Then, by the Pythagorean theorem,

$$OP^2 + PB^2 = OB^2 \Rightarrow OP^2 + 144 = 225 \Rightarrow$$
$$OP^2 = 81 \Rightarrow OP = 9$$

Since all radii are equal, $OQ = 9 \Rightarrow QB = 15 - 9 =$ **6**.

Exercises on Circles

Multiple-Choice Questions

1. What is the circumference of a circle whose area is 100π?

   (A) 10 (B) 20 (C) 10π (D) 20π (E) 25π

2. What is the area of a circle whose circumference is π?

   (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) π (D) 2π (E) 4π

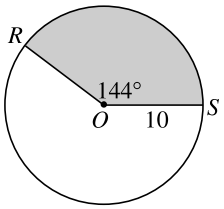
3. What is the area of a circle that is inscribed in a square of area 2?

   (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) π (D) $\pi\sqrt{2}$ (E) 2π

4. A square of area 2 is inscribed in a circle. What is the area of the circle?

   (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) π (D) $\pi\sqrt{2}$ (E) 2π

Questions 5 and 6 refer to the following figure.

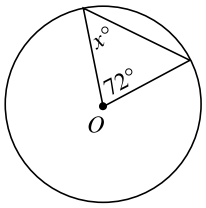


5. What is the length of arc RS?

   (A) 8 (B) 20 (C) 8π (D) 20π (E) 40π

6. What is the area of the shaded sector?

   (A) 8 (B) 20 (C) 8π (D) 20π (E) 40π



7. In the figure above, what is the value of x ?

   (A) 30 (B) 36 (C) 45 (D) 54 (E) 60

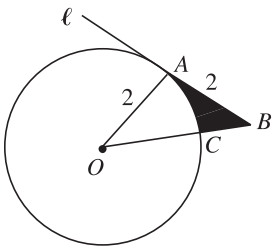
8. If A is the area and C the circumference of a circle, which of the following is an expression for A in terms of C?

   (A) $\frac{C^2}{4\pi}$ (B) $\frac{C^2}{4\pi^2}$ (C) $2C\sqrt{\pi}$ (D) $2\,C^2\sqrt{\pi}$

   (E) $\frac{C^2\sqrt{\pi}}{4}$

9. What is the area of a circle whose radius is the diagonal of a square whose area is 4?

   (A) 2π (B) $2\pi\sqrt{2}$ (C) 4π (D) 8π (E) 16π



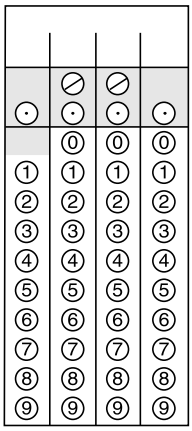
10. In the figure above, ℓ is tangent to circle O at A, and $OA = AB = 2$. What is the area of the shaded region?

    (A) $\frac{1}{2}\pi$ (B) $4 - \frac{1}{2}\pi$ (C) $2 - \frac{1}{2}\pi$

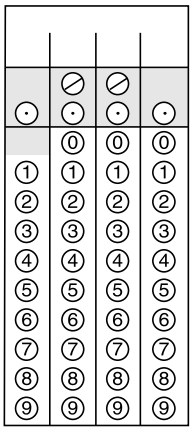
    (D) $2 - \pi$ (E) $4 - 4\pi$

Grid-in Questions

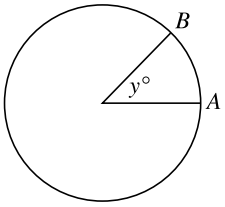
11. The circumference of a circle is $a\pi$ units, and the area of the circle is $b\pi$ square units. If $a = b$, what is the radius of the circle?

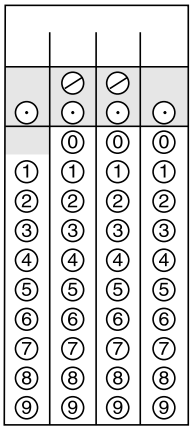


12. A 9×12 rectangle is inscribed in a circle. What is the radius of the circle?

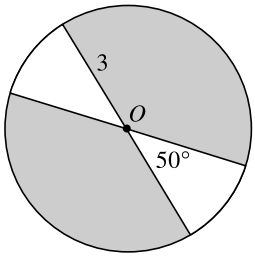


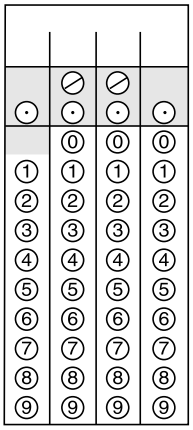
13. In the figure below, the ratio of the length of arc AB to the circumference of the circle is 2:15. What is the value of y?



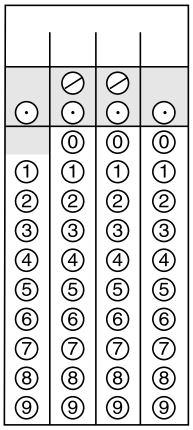


14. If the area of the shaded region is $k\pi$, what is the value of k?





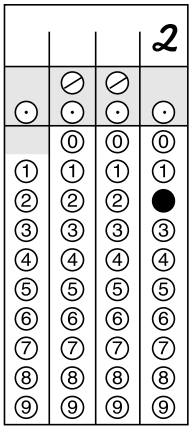
15. If line ℓ is tangent to circle O at point P, if B is a point on ℓ such that $PB = 8$, and if $OB = 10$, what is the radius of the circle?



Answer Key

1. **D**
2. **A**
3. **B**
4. **C**
5. **C**
6. **E**
7. **D**
8. **A**
9. **D**
10. **C**

11.



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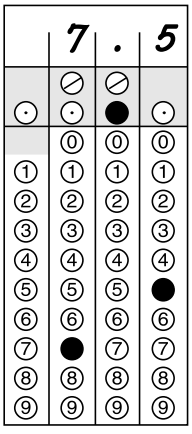

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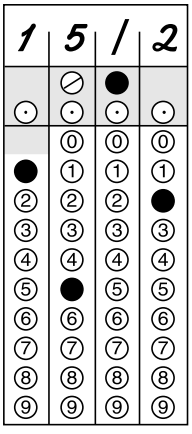

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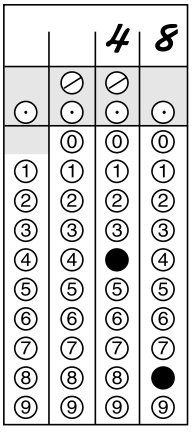
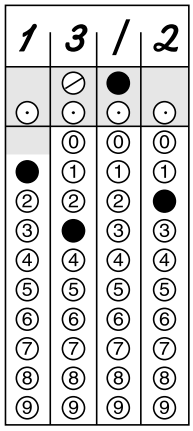
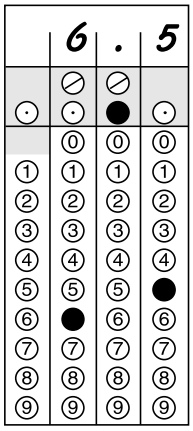
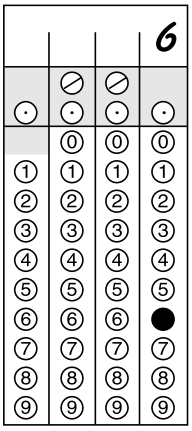


14. 

or 

15. 

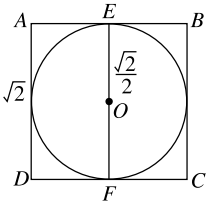
Answer Explanations

1. D. $A = \pi r^2 = 100\pi \Rightarrow r^2 = 100 \Rightarrow r = 10$.
$C = 2\pi r = 2\pi(10) = 20\pi$.

2. A. $C = 2\pi r = \pi \Rightarrow 2r = 1 \Rightarrow r = \frac{1}{2}$.

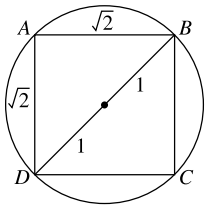
$A = \pi r^2 = \pi\left(\frac{1}{2}\right)^2 = \frac{1}{4}\pi = \frac{\pi}{4}$.

3. B. Draw a diagram. Since the area of square $ABCD$ is 2, $AD = \sqrt{2}$. Then, diameter $EF = \sqrt{2}$ and radius $OE = \frac{\sqrt{2}}{2}$, so



$$\text{area} = \pi\left(\frac{\sqrt{2}}{2}\right)^2 = \frac{2}{4}\pi = \frac{\pi}{2}.$$

4. C. Draw a diagram. Since the area of square $ABCD$ is 2, $AD = \sqrt{2}$. Then diagonal $BD = \sqrt{2} \times \sqrt{2} = 2$. But $\overline{BD}$ is also a diameter of the circle, so the diameter is 2 and the radius is 1. Therefore, the area is $\pi(1)^2 = \pi$.



5. C. The length of arc $RS = \left(\frac{144}{360}\right)2\pi(10) = \left(\frac{2}{5}\right)20\pi = 8\pi$. [Note that, instead of reducing $\frac{144}{360}$, you could have used your calculator and divided: 144 ÷ 360 = 0.4, and $(0.4)(20\pi) = 8\pi$.]

6. E. The area of the shaded sector is
$$\left(\frac{144}{360}\right)\pi(10)^2 = \left(\frac{2}{5}\right)100\pi = 40\pi.$$

7. D. The triangle is isosceles, so the third (unmarked) angle is also x:
$180 = 72 + 2x \Rightarrow 2x = 108 \Rightarrow x = 54$.

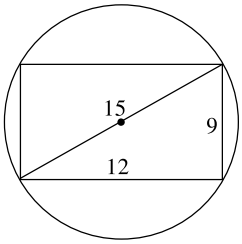
8. A. $C = 2\pi r \Rightarrow r = \frac{C}{2\pi} \Rightarrow A = \pi\left(\frac{C}{2\pi}\right)^2 = \pi\left(\frac{C^2}{4\pi^2}\right) = \frac{C^2}{4\pi}$.

9. D. If the area of the square is 4, each side is 2, and the length of a diagonal is $2\sqrt{2}$.
The area of a circle whose radius is $2\sqrt{2}$ is $\pi(2\sqrt{2})^2 = 8\pi$.

10. C. Since ℓ is tangent to circle O at A, $\overline{OA} \perp \ell$ and $\triangle OAB$ is an isosceles right triangle. Then $m\angle O = 45$.
The area of the shaded region is the area of $\triangle OAB$ minus the area of sector OAC.
The area of $\triangle OAB$ is $\frac{1}{2}(2)(2) = 2$. Since the area of the circle is $\pi(2^2) = 4\pi$, the area of sector OAC is $\frac{45}{360} = \frac{1}{8}$ of 4π, or $\frac{1}{2}\pi$.
Finally, the area of the shaded region is $2 - \frac{1}{2}\pi$.

11. (2) Since $a = b$, then $C = a\pi = b\pi = A$, so
$2\pi r = \pi r^2 \Rightarrow 2r = r^2 \Rightarrow r = 2$.

12. $\left(7.5 \text{ or } \frac{15}{2}\right)$ Draw a diagram. By the Pythagorean theorem (or by recognizing a $3x$-$4x$-$5x$ triangle with $x = 3$), the length of diagonal $\overline{AC}$ is 15. But $\overline{AC}$ is also a diameter of the circle, so the diameter is 15 and the radius is 7.5 or $\frac{15}{2}$.

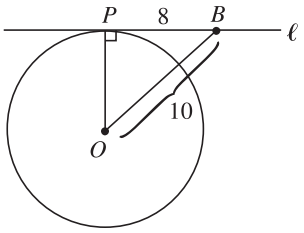


13. (48) Since arc AB is $\frac{2}{15}$ of the circumference, y is $\frac{2}{15} \times 360 = 48$.

14. $\left(\frac{13}{2} \text{ or } 6.5\right)$ The area of the circle is 9π; and since the white region is $\frac{100}{360} = \frac{5}{18}$ of the circle, the shaded region is $\frac{13}{18}$ of it:

$$\frac{13}{18} \times 9\pi = \frac{13}{2}\pi \text{ or } 6.5\pi.$$

15. (6) Draw a diagram and label it.



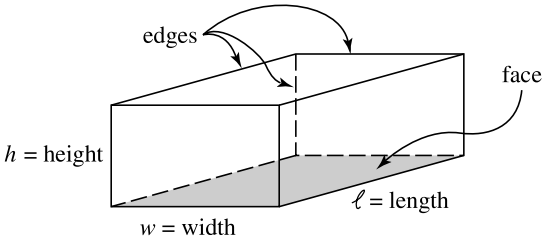
Since radius $\overline{OP}$ is perpendicular to ℓ, $\triangle OPB$ is a right triangle.
By the Pythagorean theorem,

$$OP^2 + 8^2 = 10^2 \Rightarrow OP^2 + 64 = 100 \Rightarrow$$
$$OP^2 = 36 \Rightarrow OP = 6.$$

12-M SOLID GEOMETRY

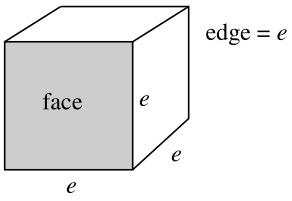
There is very little solid geometry on the SAT. Basically, all you need to know are the formulas for the volumes and surface areas of rectangular solids (including cubes) and cylinders.

A ***rectangular solid*** or ***box*** is a solid formed by six rectangles, called ***faces***. The sides of the rectangles are called ***edges***. As shown in the diagram that follows, the edges are the ***length***, ***width***, and ***height***.



RECTANGULAR SOLID

A ***cube*** is a rectangular solid in which the length, width, and height are equal, so that all the edges are the same length.



CUBE

The ***volume*** of a solid, which is the amount of space it occupies, is measured in ***cubic units***. One cubic unit is the amount of space occupied by a cube all of whose edges are 1 unit long. In the figure above, if the length of each edge of the cube is 1 inch, the area of each face is 1 square inch, and the volume of the cube is 1 cubic inch.

Key Fact M1

The formula for the volume of a rectangular solid is $V = \ell wh$.

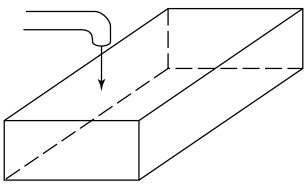
In a cube, all the edges are equal. Therefore, if e is the length of an edge, the formula for the volume is $V = e^3$.

KEY FACT M1 is one of the facts provided in the "Reference Information" at the beginning of each math section.

Example 1.

The base of a rectangular tank is 2 feet wide and 4 feet long; the height of the tank is 20 inches. If water is pouring into the tank at the rate of 2 cubic inches per second, how many hours will be required to fill the tank?

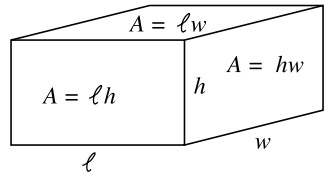
Solution. Draw a diagram. Change all units to inches. Then the volume of the tank is $24 \times 48 \times 20 = 23{,}040$ cubic inches. At 2 cubic inches per second:

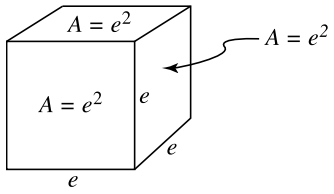


$$\text{required time} = \frac{23{,}040}{2} = 11{,}520 \text{ seconds} = \frac{11{,}520}{60} =$$

$$192 \text{ minutes} = \frac{192}{60} = \mathbf{3.2} \text{ hours.}$$

The ***surface area*** of a rectangular solid is the sum of the areas of the six faces. Since the top and bottom faces are equal, the front and back faces are equal, and the left and right faces are equal, you can calculate the area of one face from each pair and then double the sum. In a cube, each of the six faces has the same area.



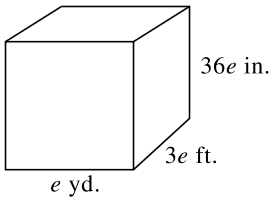


Key Fact M2

The formula for the surface area of a rectangular solid is $A = 2(\ell w + \ell h + wh)$. The formula for the surface area of a cube is $A = 6e^2$.

Example 2.

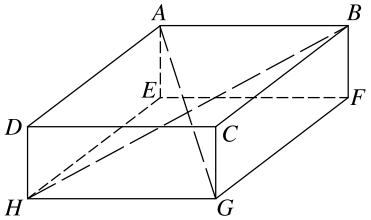
The volume of a cube is v cubic yards, and its surface area is a square feet. If $v = a$, what is the length, in inches, of each edge?



Solution. Draw a diagram. If e is the length of the edge in yards, then $3e$ is the length in feet, and $36e$ is the length in inches. Therefore, $v = e^3$ and $a = 6(3e)^2 = 6(9e^2) = 54e^2$.

Since $v = a$, $e^3 = 54e^2 \Rightarrow e = 54$; the length of each edge is $36(54) = \mathbf{1944}$ inches.

A ***diagonal*** of a box is a line segment joining a vertex on the top of the box to the opposite vertex on the bottom. A box has four diagonals, all the same length. In the box below they are line segments $\overline{AG}$, $\overline{BH}$, $\overline{CE}$, and $\overline{DF}$.



Key Fact M3

A diagonal of a box is the longest line segment that can be drawn between two points on the box.

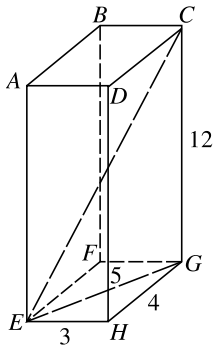
Key Fact M4

If the dimensions of a box are ℓ, w, and h, and if d is the length of a diagonal, then $d^2 = \ell^2 + w^2 + h^2$.

For example, in the box below:

$$d^2 = 3^2 + 4^2 + 12^2 = 9 + 16 + 144 = 169 \Rightarrow d = 13.$$

This formula is really just an extended Pythagorean theorem. $\overline{EG}$ is the diagonal of rectangular base $EFGH$. Since the sides of the base are 3 and 4, EG is 5. Now, $\triangle CGE$ is a right triangle whose legs are 12 and 5, so diagonal $\overline{CE}$ is 13. (The only reason not to use the Pythagorean theorem is that these triangles are so familiar.)



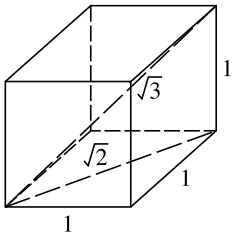
Example 3.

What is the length of a diagonal of a cube whose sides are 1?

Solution. Use the formula:

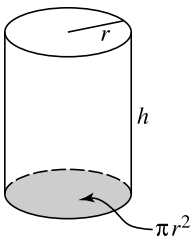
$$d^2 = 1^2 + 1^2 + 1^2 = 3 \Rightarrow d = \sqrt{\mathbf{3}}.$$

Without the formula you would draw a diagram and label it. Since the base is a 1×1 square, its diagonal is $\sqrt{2}$. Then the diagonal of the cube is the hypotenuse of a right triangle whose legs are 1 and $\sqrt{2}$, so



$$d^2 = 1^2 + (\sqrt{2})^2 = 1 + 2 = 3, \quad \text{and} \quad d = \sqrt{\mathbf{3}}.$$

A ***cylinder*** is similar to a rectangular solid except that the base is a circle instead of a rectangle. The volume of a cylinder is the area of its circular base (πr^2) times its height (h). The surface area of a cylinder depends on whether you are envisioning a tube, such as a straw, without a top or bottom, or a can, which has both a top and a bottom.



Key Fact M5

- **The volume, V, of a cylinder whose circular base has radius r and whose height is h is the area of the base times the height:**

$$V = \pi r^2 h.$$

- **The surface area, A, of the side of the cylinder is the circumference of the circular base times the height:**

$$A = 2\pi rh.$$

The formula for the volume of a cylinder is one of the facts provided in the "Reference Information" at the beginning of each math section.

Example 4.

The volume of a cube and the volume of a cylinder are equal. If the edge of the cube and the radius of the cylinder are each 6, which of the following is the best approximation of the height of the cube?

(A) 1 (B) 2 (C) 3 (D) 6 (E) 12

Solution. The volume of the cube is $6^3 = 216$. The volume of the cylinder is $\pi(6^2)h = 36\pi h$. Then

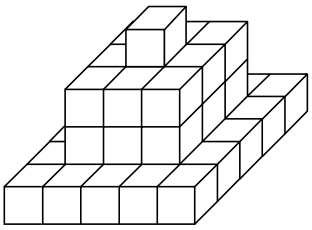
$$216 = 36\pi h \Rightarrow \pi h = 6 \Rightarrow h = \frac{6}{\pi}.$$

Since π is approximately 3, h is approximately **2 (B).**

You now know the only formulas you will need. Any other solid geometry questions that may appear on the SAT will require you to visualize a situation and reason it out, rather than to apply a formula.

Example 5.

How many small blocks are needed to construct the tower in the figure at the right?



Solution. You need to "see" the answer. The top level consists of 1 block, the second and third levels consist of 9 blocks each, and the bottom layer consists of 25 blocks. The total is 1 + 9 + 9 + 25 = **44.**

Exercises on Solid Geometry

Multiple-Choice Questions

1. What is the volume of a cube whose surface area is 150?

   (A) 25 (B) 100 (C) 125 (D) 1000 (E) 15,625

2. What is the surface area of a cube whose volume is 64?

   (A) 16 (B) 64 (C) 96 (D) 128 (E) 384

3. A solid metal cube of side 3 inches is placed in a rectangular tank whose length, width, and height are 3, 4, and 5 inches, respectively. What is the volume, in cubic units, of water that the tank can now hold?

   (A) 20 (B) 27 (C) 33 (D) 48 (E) 60

4. The height, h, of a cylinder is equal to the edge of a cube. If the cylinder and the cube have the same volume, what is the radius of the cylinder?

   (A) $\frac{h}{\sqrt{\pi}}$ (B) $h\sqrt{\pi}$ (C) $\frac{\sqrt{\pi}}{h}$ (D) $\frac{h^2}{\pi}$ (E) πh^2

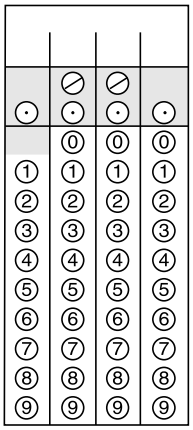
5. If the height of a cylinder is 4 times its circumference, what is the volume of the cylinder in terms of its circumference, C?

   (A) $\frac{C^3}{\pi}$ (B) $\frac{2C^3}{\pi}$ (C) $\frac{2C^2}{\pi^2}$ (D) $\frac{\pi C^2}{4}$

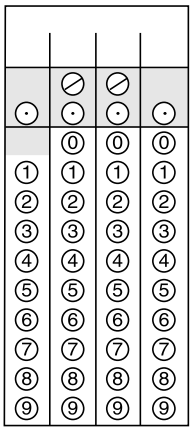
   (E) $4\pi C^3$

Grid-in Questions

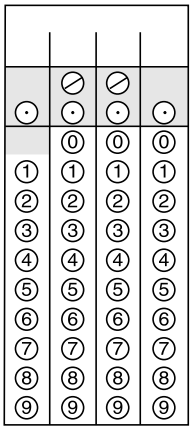
6. The sum of the lengths of all the edges of a cube is 6 centimeters. What is the volume, in cubic centimeters, of the cube?



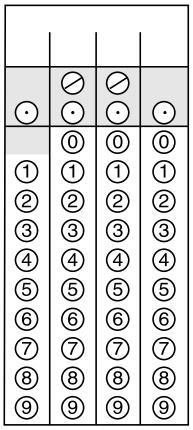
7. A 5-foot-long cylindrical pipe has an inner diameter of 6 feet and an outer diameter of 8 feet. If the total surface area (inside and out, including the ends) is $k\pi$, what is the value of k?



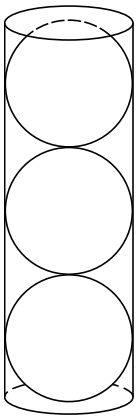
8. What is the number of cubic inches in 1 cubic foot?

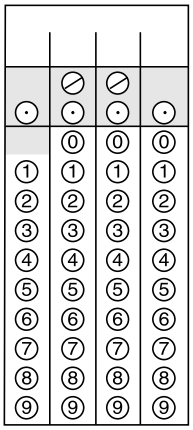


9. A rectangular tank has a base that is 10 centimeters by 5 centimeters and a height of 20 centimeters. If the tank is half full of water, by how many centimeters will the water level rise if 325 cubic centimeters are poured into the tank?



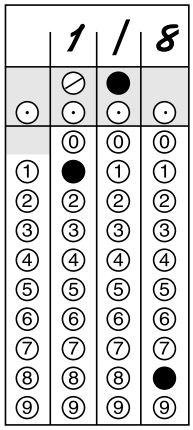
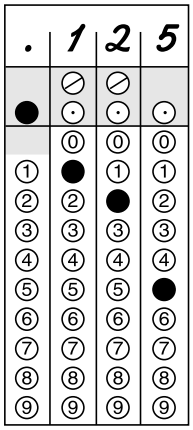
10. Three identical balls fit snugly into a cylindrical can: the radius of the spheres equals the radius of the can, and the balls just touch the bottom and the top of the can. If the formula for the volume of a sphere is $V = \frac{4}{3}\pi r^3$, what fraction of the volume of the can is taken up by the balls?

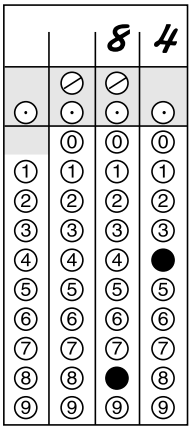


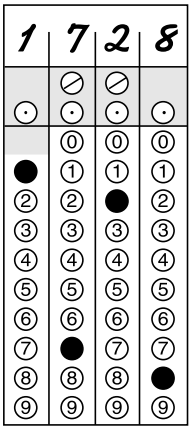


Answer Key

1. **C** 2. **C** 3. **C** 4. **A** 5. **A**

6.  or 

7. 

8. 

9.

	6	.	5
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⊙	⊙	●	⊙
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	●
6	●	6	6
7	7	7	7
8	8	8	8
9	9	9	9

10.

	2	/	3
	⊘	●	
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2	●	2	2
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4	4	4	4
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9	9	9	9

or

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3		3	3	3
4		4	4	4
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6		●	●	●
7		7	7	7
8		8	8	8
9		9	9	9

or

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2		2	2	2
3		3	3	3
4		4	4	4
5		5	5	5
6		●	●	6
7		7	7	●
8		8	8	8
9		9	9	9

Answer Explanations

1. C. Since the surface area is 150, each of the six faces of the cube is a square whose area is $150 \div 6 = 25$. Then, each edge is 5, and the volume is $5^3 = 125$.

2. C. Since the volume of the cube is 64, then $e^3 = 64 \Rightarrow e = 4$. The surface area is $6e^2 = 6 \times 16 = 96$.

3. C. The volume of the tank is $3 \times 4 \times 5 = 60$ cubic units, but the solid cube is taking up $3^3 = 27$ cubic units. Therefore, the tank can hold $60 - 27 = 33$ cubic units of water.

4. A. Since the volumes are equal, $\pi r^2 h = e^3 = h^3$. Therefore,

$$\pi r^2 = h^2 \Rightarrow r^2 = \frac{h^2}{\pi} \Rightarrow r = \frac{h}{\sqrt{\pi}}.$$

5. A. Since $V = \pi r^2 h$, you need to express r and h in terms of C. It is given that $h = 4C$; and since $C = 2\pi r$, then $r = \dfrac{C}{2\pi}$. Therefore,

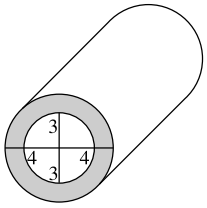
$$V = \pi\left(\frac{C}{2\pi}\right)^2 (4C) = \pi\left(\frac{C^2}{4\pi^2}\right)(4C) = \frac{C^3}{\pi}.$$

6. $\left(\frac{1}{8}\text{ or }.125\right)$ Since a cube has 12 edges:

$$12e = 6 \Rightarrow e = \frac{1}{2}.$$

Therefore: $V = e^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ or .125.

7. (84) Draw a diagram and label it. Since the surface of a cylinder is given by $A = 2\pi rh$, the area of the exterior is $2\pi(4)(5) = 40\pi$, and the area of the interior is $2\pi(3)(5) = 30\pi$. The area of *each* shaded end is the area of the outer circle minus the area of the inner circle: $16\pi - 9\pi = 7\pi$, so

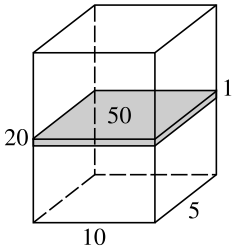


total surface area =
$40\pi + 30\pi + 7\pi + 7\pi = 84\pi \Rightarrow k = 84$.

8. (1728) The volume of a cube whose edges are 1 foot can be expressed in either of two ways:

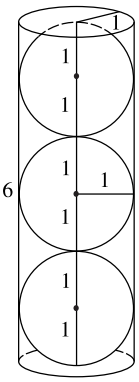
$(1 \text{ foot})^3 = 1$ cubic foot or
$(12 \text{ inches})^3 = 1728$ cubic inches.

9. (6.5) Draw a diagram. Since the area of the base is $5 \times 10 = 50$ square centimeters, each 1 centimeter of depth has a volume of 50 cubic centimeters. Therefore, 325 cubic centimeters will raise the water level $325 \div 50 = 6.5$ centimeters.



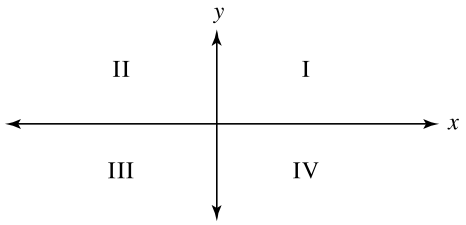
(Note that the fact that the tank was half full was not used, except to be sure that the tank didn't overflow. Since the tank was half full, the water was 10 centimeters deep, and the water level could rise by 6.5 centimeters. Had the tank been three-fourths full, the water would have been 15 centimeters deep, and the extra water would have caused the level to rise 5 centimeters, filling the tank; the rest of the water would have spilled out.)

10. $\left(\frac{2}{3}\text{ or .666 or .667}\right)$ To avoid using r, assume that the radii of the spheres and the can are 1. Then the volume of each ball is $\frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$, and the total volume of the three balls is $3\left(\frac{4}{3}\pi\right) = 4\pi$. Since the volume of the can is $\pi(1)^2(6) = 6\pi$, the balls take up $\dfrac{4\pi}{6\pi} = \dfrac{2}{3}$ of the can. Grid in $\frac{2}{3}$ or .666 or .667.



12-N COORDINATE GEOMETRY

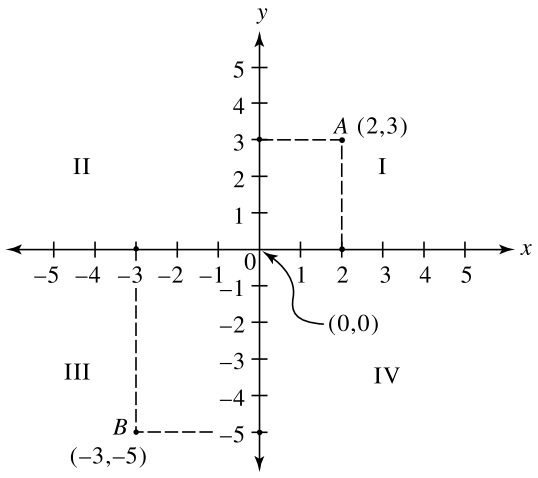
The coordinate plane is formed by two perpendicular number lines called the ***x-axis*** and ***y-axis***, which intersect at the ***origin***. The axes divide the plane into four ***quadrants***, labeled, in counterclockwise order, I, II, III, and IV.



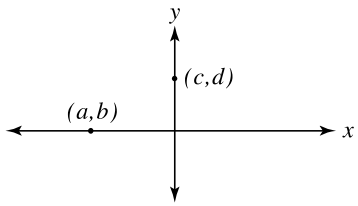
Each point in the plane is assigned two numbers, an ***x-coordinate*** and a ***y-coordinate***, which are written as an ordered pair, **(*x*, *y*)**.

- Points to the right of the *y*-axis have positive *x*-coordinates, and those to the left have negative *x*-coordinates.
- Points above the *x*-axis have positive *y*-coordinates, and those below it have negative *y*-coordinates.
- If a point is on the *x*-axis, its *y*-coordinate is 0.
- If a point is on the *y*-axis, its *x*-coordinate is 0.

For example, point *A* in the figure below is labeled (2, 3), since it is 2 units to the right of the *y*-axis and 3 units above the *x*-axis. Similarly, point *B*(–3, –5) is in Quadrant III, 3 units to the left of the *y*-axis and 5 units below the *x*-axis.



Example 1.

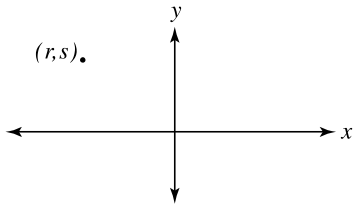


In the figure above, all of the following are equal EXCEPT

(A) *ab* (B) *ac* (C) *ad* (D) *bc* (E) *bd*

Solution. Since (*a, b*) lies on the *x*-axis, *b* = 0. Since (*c, d*) lies on the *y*-axis, *c* = 0. Each of the choices is equal to 0 except ***ad*** **(C).**

Example 2.



In the figure above, which of the following *must* be true?

I. $rs < 0$ II. $r < s$ III. $r + s = 0$

(A) None (B) I only (C) II only
(D) I and II only (E) I, II, and III

Solution. Since (*r, s*) is in Quadrant II, *r* is negative and *s* is positive. Then $rs < 0$ (I is true) and $r < s$ (II is true). Although $r + s$ could be equal to 0, it does not have to equal 0 (III is false). Only **I** and **II** must be true **(D).**

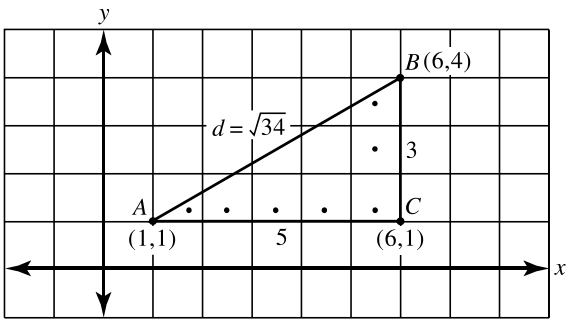
Often a question requires you to calculate the distance between two points. This task is easiest when the points lie on the same horizontal or vertical line.

Key Fact N1

- **All the points on a horizontal line have the same *y*-coordinate. To find the distance between them, subtract their *x*-coordinates.**
- **All the points on a vertical line have the same *x*-coordinate. To find the distance between them, subtract their *y*-coordinates.**

Helpful Hint

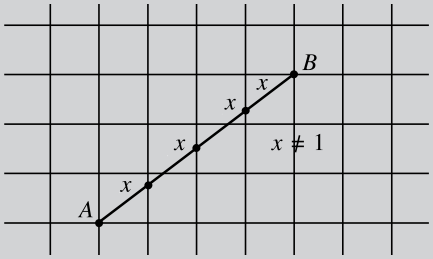
If the points have been plotted on a graph, you can find the distance between them by counting boxes.



In the graph, the distance from A to C is 6 – 1 = 5. The distance from B to C is 4 – 1 = 3.

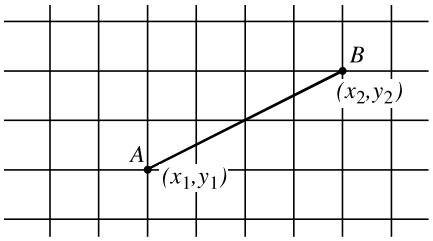
It is a little harder, but not much, to find the distance between two points that are not on the same horizontal or vertical line; just use the Pythagorean theorem. For example, in the preceding graph, if d represents the distance from A to B, $d^2 = 5^2 + 3^2 = 25 + 9 = 34 \Rightarrow d = \sqrt{34}$.

CAUTION: You *cannot* count boxes unless the points are on the same horizontal or vertical line. The distance between A and B is 5, not 4.



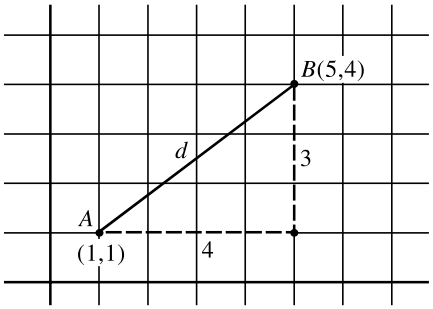
Key Fact N2

The distance, d, between two points, $A(x_1, y_1)$ and $B(x_2, y_2)$, can be calculated using the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

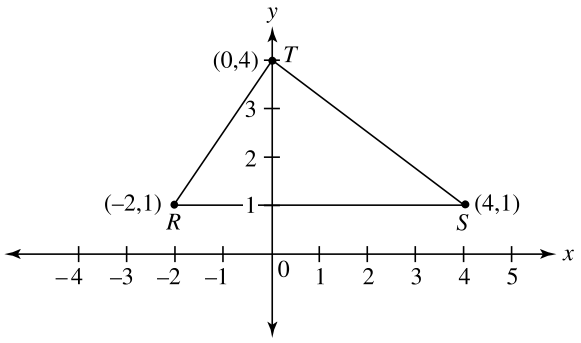


Helpful Hint

The distance formula is nothing more than the Pythagorean theorem. If you ever forget the formula, and you need the distance between two points that do not lie on the same horizontal or vertical line, do as follows: create a right triangle by drawing a horizontal line through one of the points and a vertical line through the other, and then use the Pythagorean theorem.



Examples 3 and 4 refer to the triangle in the following figure.



Example 3.

What is the area of $\triangle RST$?

(A) 6 (B) 9 (C) 12 (D) 15 (E) 18

Solution. $R(-2, 1)$ and $S(4, 1)$ lie on the same horizontal line, so $RS = 4 - (-2) = 6$. Let that be the base of the triangle. Then the height is the distance along the vertical line from T to RS: 4 – 1 = 3. The area is $\frac{1}{2}(6)(3) =$ **9 (B).**

Example 4.

What is the perimeter of $\triangle RST$?

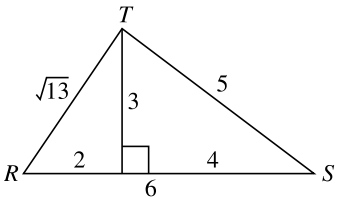
(A) 13 (B) 14 (C) 16 (D) $11 + \sqrt{13}$ (E) $11 + \sqrt{61}$

Solution. The perimeter is $RS + ST + RT$. From the solution to Example 3, you know that $RS = 6$. Also, $ST = 5$, since it is the hypotenuse of a 3-4-5 right triangle. To calculate RT, use either the distance formula:

$$\sqrt{(-2-0)^2 + (1-4)^2} = \sqrt{(-2)^2 + (-3)^2} = \sqrt{4+9} = \sqrt{13}$$

or the Pythagorean theorem:

$$RT^2 = 2^2 + 3^2 = 4 + 9 = 13 \Rightarrow RT = \sqrt{13}.$$



Then the perimeter is $6 + 5 + \sqrt{13} = \mathbf{11 + \sqrt{13}}$ **(D).**

Key Fact N3

If $P\ (x_1, y_1)$ and $Q\ (x_2, y_2)$ are any two points, then the midpoint, M, of segment $\overline{PQ}$ is the point whose coordinates are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Example 5.

A (2, –3) and B (8, 5) are the endpoints of a diameter of a circle. What are the coordinates of the center of the circle?

(A) (3, 1) (B) (3, 4) (C) (5, 1) (D) (5, 4) (E) (10, 2)

Solution. The center of a circle is the midpoint of any diameter. Therefore, the coordinates are $\left(\frac{2+8}{2}, \frac{-3+5}{2}\right) = \left(\frac{10}{2}, \frac{2}{2}\right) =$ **(5, 1) (C).**

The ***slope*** of a line is a number that indicates how steep the line is.

Key Fact N4

- **Vertical lines *do not have slopes*.**
- **To find the slope of any other line, proceed as follows:**
  1. **Choose any two points, $A(x_1, y_1)$ and $B(x_2, y_2)$, on the line.**
  2. **Take the differences of the y-coordinates, $y_2 - y_1$, and the x-coordinates, $x_2 - x_1$.**
  3. **Divide: slope $= \frac{y_2 - y_1}{x_2 - x_1}$.**

We will illustrate the next KEY FACT by using this formula to calculate the slopes of RS, RT, and ST from Example 3: $R(-2, 1)$, $S(4, 1)$, $T(0, 4)$.

Key Fact N5

- **The slope of any horizontal line is 0:**

$$\text{slope of } \overline{RS} = \frac{1-1}{4-(-2)} = \frac{0}{6} = 0.$$

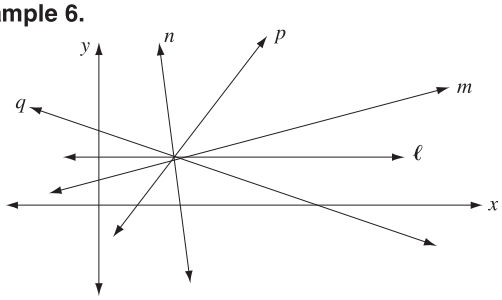
- **The slope of any line that goes up as you move from left to right is positive:**

$$\text{slope of } \overline{RT} = \frac{4-1}{0-(-2)} = \frac{3}{2}.$$

- **The slope of any line that goes down as you move from left to right is negative:**

$$\text{slope of } \overline{ST} = \frac{1-4}{4-0} = -\frac{3}{4}.$$

Example 6.



In the figure above, which line has the greatest slope?

(A) ℓ (B) m (C) n (D) p (E) q

Solution. Since the slope of line ℓ is 0 and the slopes of lines n and q are negative, eliminate choices A, C, and E. Lines m and p have positive slopes, but line p is steeper. **(D)**

The next key fact concerns the relationships between the slopes of parallel and perpendicular lines.

Key Fact N6

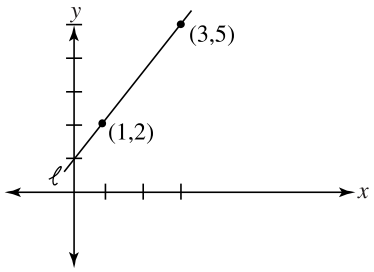
- **If two nonvertical lines are parallel, their slopes are equal.**
- **If two nonvertical lines are perpendicular, the product of their slopes is –1.**

If the product of two numbers, r and s, is –1, then

$$rs = -1 \Rightarrow r = -\frac{1}{s}.$$

Therefore, another way to express the second part of KEY FACT N6 is to say that, if two nonvertical lines are perpendicular, the slope of one is the negative reciprocal of the slope of the other.

Example 7.



In the figure above, line ℓ passes through points (1, 2) and (3, 5). Line m (not shown) is perpendicular to ℓ. What is the slope of line m?

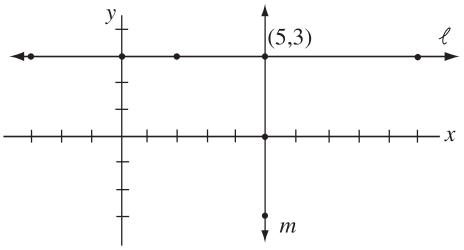
(A) $-\frac{3}{2}$ (B) $-\frac{2}{3}$ (C) 0 (D) $\frac{2}{3}$ (E) $\frac{3}{2}$

Solution. First use the slope formula to calculate the slope of line ℓ: $\frac{5-2}{3-1} = \frac{3}{2}$.

Then the slope of line m is the negative reciprocal of $\frac{3}{2}$, which is $-\frac{2}{3}$ **(B).**

Note that you can see from the diagram in Example 7 that the slope of ℓ is positive. If you sketch any line perpendicular to ℓ, you can see that its slope is negative. So immediately you know the answer must be A or B.

Every line that is drawn in the coordinate plane has an equation. All the points on a horizontal line have the same *y*-coordinate. For example, in the figure below, horizontal line ℓ passes through (–3, 3), (0, 3), (2, 3), (5, 3), and (10, 3).



The equation of line ℓ is $y = 3$.

Similarly, every point on vertical line *m* has an *x*-coordinate equal to 5, and the equation of *m* is $x = 5$.

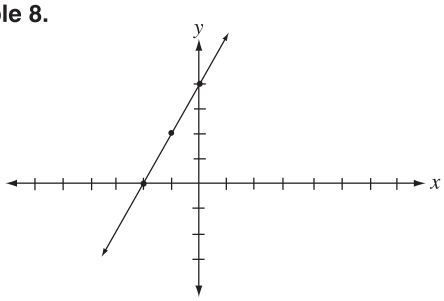
Every other line in the coordinate plane has an equation that can be written in the form $y = mx + b$, where *m* is the slope of the line and *b* is the ***y-intercept***—the *y*-coordinate of the point where the line crosses the *y*-axis. These facts are summarized in KEY FACT N7.

Key Fact N7

- **For any real number *a*: $x = a$ is the equation of the vertical line that crosses the *x*-axis at (*a*, 0).**
- **For any real number *b*: $y = b$ is the equation of the horizontal line that crosses the *y*-axis at (0, *b*).**
- **For any real numbers *b* and *m*: $y = mx + b$ is the equation of the line that crosses the *y*-axis at (0, *b*) and whose slope is *m*.**

On the SAT, you won't have to graph a line, but you may have to recognize the graph of a line. In a multiple-choice question, you may be given the graph of a line and asked which of the five choices is the equation of that line; or you may be given the equation of a line and asked which of the five choices is the correct graph.

Example 8.



Which of the following is the equation of the line in the figure above?

(A) $y = 2x + 4$ (B) $y = \frac{1}{2}x + 4$ (C) $y = 2x - 2$

(D) $y = \frac{1}{2}x - 4$ (E) $y = 4x + 2$

There are two different ways to handle this question.

Solution 1. Since the line is neither horizontal nor vertical, its equation has the form $y = mx + b$. Since the line crosses the *y*-axis at 4, $b = 4$. Also, since the line passes through (–2, 0) and (0, 4), its slope is $\frac{4-0}{0-(-2)} = \frac{4}{2} = 2$.

So $m = 2$, and the equation is ***y* = 2*x* + 4 (A).**

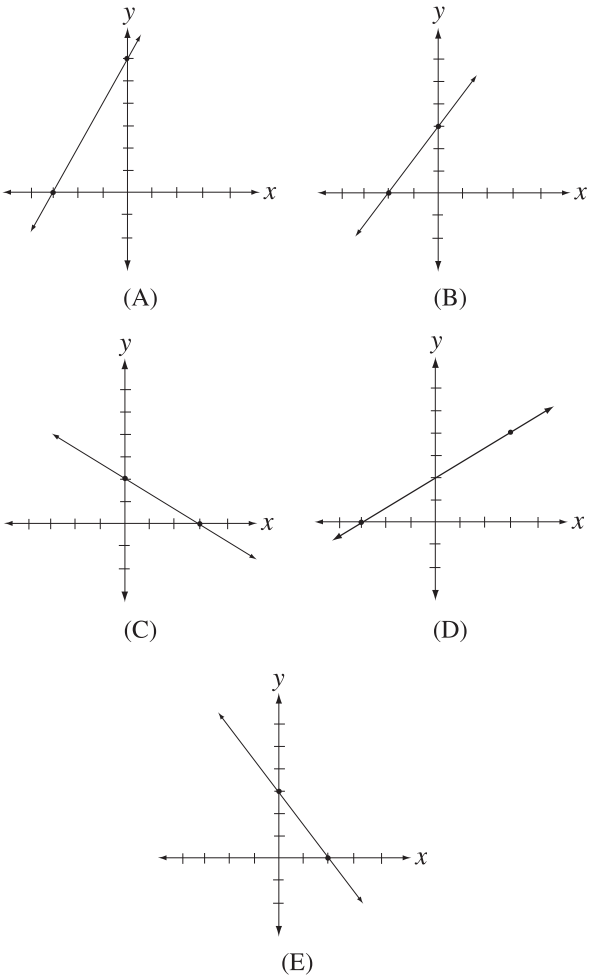
Solution 2. Test some points. Since the line passes through (0, 4), $y = 4$ when $x = 0$. Plug in 0 for *x* in the five choices; only in A and B does *y* equal 4. The line also passes through (–2, 0), so when $x = -2$, $y = 0$.

Replace *x* by –2 in choices A and B.

- $2(-2) + 4 = -4 + 4 = 0$, so A works.
- $\frac{1}{2}(-2) + 4 = -1 + 4 = 3$, so B does not work.

Example 9.

Which of the following is the graph of $3y = 2x + 6$?



Solution 1. Express $3y = 2x + 6$ in standard form by dividing each term by 3: $y = \frac{2}{3}x + 2$. From this equation you see that the y-intercept is 2. Eliminate choices A, B, and E. Since the slope $\frac{2}{3}$ is positive, eliminate C. The answer must be **D.**

Solution 2. Test some points.

- When $x = 0$, $3y = 2(0) + 6 = 6 \Rightarrow y = 2$, so (0, 2) is a point on the graph.
- When $x = 3$, $3y = 2(3) + 6 = 12 \Rightarrow y = 4$, so (3, 4) is on the graph.

Only choice **D** passes through both (0, 2) and (3, 4).

Exercises on Coordinate Geometry

Multiple-Choice Questions

1. If $A(-1, 1)$ and $B(3, -1)$ are the endpoints of one side of square $ABCD$, what is the area of the square?
   (A) 12 (B) 16 (C) 20 (D) 25 (E) 36

2. If $P(2, 1)$ and $Q(8, 1)$ are two of the vertices of a rectangle, which of the following *cannot* be another of the vertices?
   (A) (2, 8) (B) (8, 2) (C) (2, –8) (D) (–2, 8) (E) (8, 8)

3. A circle whose center is at (6,8) passes through the origin. Which of the following points is NOT on the circle?
   (A) (12, 0) (B) (6, –2) (C) (16, 8) (D) (–2, 12) (E) (–4, 8)

4. What is the slope of the line that passes through (a, b) and $\left(\frac{1}{a}, b\right)$?
   (A) 0 (B) $\frac{1}{b}$ (C) $\frac{1-a^2}{a}$ (D) $\frac{a^2-1}{a}$ (E) Undefined

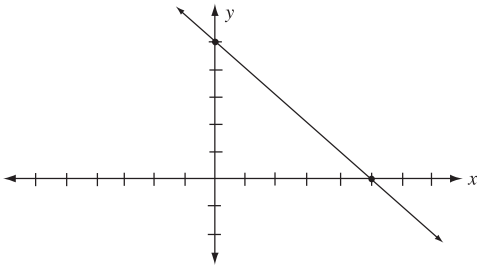
5. If $c \neq 0$ and the slope of the line passing through $(-c, c)$ and $(3c, a)$ is 1, which of the following is an expression for a in terms of c?
   (A) $-3c$ (B) $-\frac{c}{3}$ (C) $2c$ (D) $3c$ (E) $5c$

6. What is the slope of the line that passes through (3, 2) and is parallel to the line that passes through (–2, 3) and (2, –3)?
   (A) $-\frac{3}{2}$ (B) $-\frac{2}{3}$ (C) $\frac{2}{3}$ (D) 1 (E) $\frac{3}{2}$

7. What is the slope of the line that passes through (3, 2) and is perpendicular to the line that passes through (–2, 3) and (2, –3)?
   (A) $-\frac{3}{2}$ (B) $-\frac{2}{3}$ (C) $\frac{2}{3}$ (D) 1 (E) $\frac{3}{2}$

8. Line ℓ is tangent to the circle whose center is at (3, 2). If the point of tangency is (6, 6), what is the slope of line ℓ?
   (A) $-\frac{4}{3}$ (B) $-\frac{3}{4}$ (C) 0 (D) $\frac{3}{4}$ (E) $\frac{4}{3}$

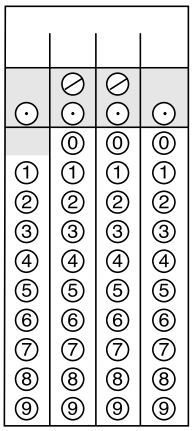
9. What is the equation of the line that passes through (4, –4) and (4, 4)?
   (A) $x = 4$ (B) $y = 4$ (C) $y = 4x$ (D) $y = 4x + 4$ (E) $y = 4x - 4$



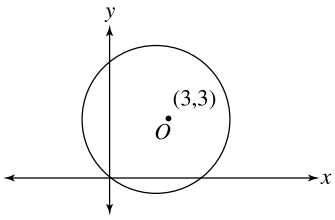
10. In the figure above, what is the equation of the line that crosses the y-axis at (0, 5) and crosses the x-axis at (5, 0)?
    (A) $x = 5$ (B) $y = 5$ (C) $y = x + 5$ (D) $y = x - 5$ (E) $y = -x + 5$

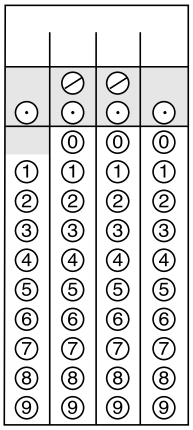
Grid-in Questions

11. If the coordinates of $\triangle RST$ are $R(0, 0)$, $S(7, 0)$, and $T(2, 5)$, what is the sum of the slopes of the three sides of the triangle?

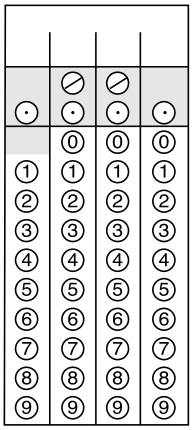


12. If the area of circle O below is $k\pi$, what is the value of k?



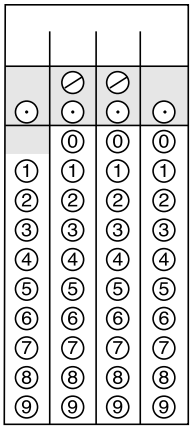


14. What is the perimeter of parallelogram $JKLM$?

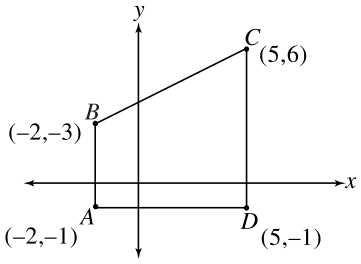


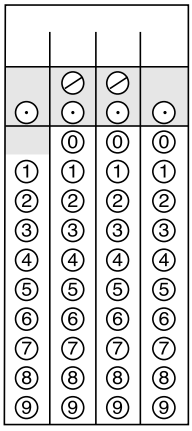
Questions 13 and 14 concern parallelogram $JKLM$, whose coordinates are $J(-5, 2)$, $K(-2, 6)$, $L(5, 6)$, $M(2, 2)$.

13. What is the area of parallelogram $JKLM$?



15. What is the area of quadrilateral $ABCD$?

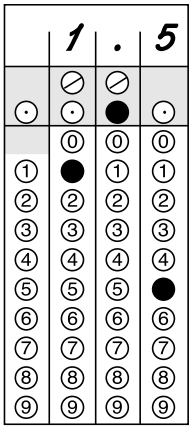




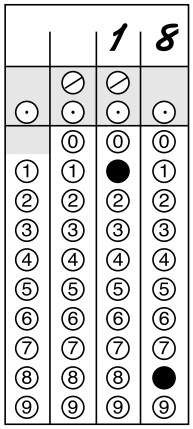
Answer Key

1. **C**
2. **D**
3. **D**
4. **A**
5. **E**
6. **A**
7. **C**
8. **B**
9. **A**
10. **E**

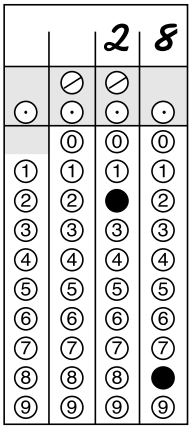
11.



12.



13.



14.

		2	4
⊙	⊘ ⊙	⊘ ⊙	⊙
	⓪	⓪	⓪
①	①	①	①
②	②	●	②
③	③	③	③
④	④	④	●
⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨

15.

3	8	.	5
⊙	⊘ ⊙	⊘ ●	⊙
	⓪	⓪	⓪
①	①	①	①
②	②	②	②
●	③	③	③
④	④	④	④
⑤	⑤	⑤	●
⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦
⑧	●	⑧	⑧
⑨	⑨	⑨	⑨

or

7	7	/	2
⊙	⊘ ⊙	● ⊙	⊙
	⓪	⓪	⓪
①	①	①	①
②	②	②	●
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥
●	●	⑦	⑦
⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨

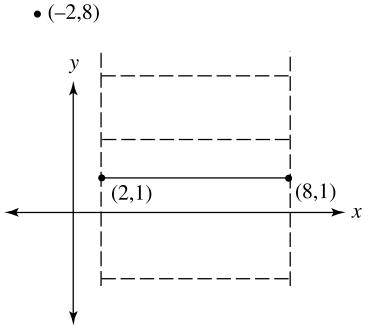
Answer Explanations

1. C. Draw a diagram and label it. The area of square $ABCD$ is s^2, where $s = AB$.
To calculate s, use the distance formula:

$$s = \sqrt{(3-(-1))^2 + (-1-1)^2} = \sqrt{4^2 + (-2)^2} = \sqrt{16+4} = \sqrt{20}.$$

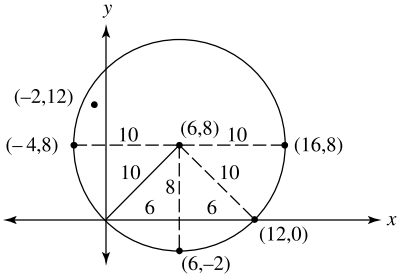
Then $s^2 = 20$.

2. D. Draw a diagram. Any point whose x-coordinate is 2 or 8 could be another vertex. Of the choices, only (–2, 8) is *not* possible.



3. D. Draw a diagram. The radius of the circle is 10 (since it's the hypotenuse of a 6-8-10 right triangle). Which of the choices is (are) 10 units from (6, 8)?
   - First, check the easy ones; E: (–4, 8) and C: (16, 8) are 10 units to the left and right of (6, 8), and B: (6, –2) is 10 units below.
   - Check A: (12, 0), which works, and D: (–2, 12), which doesn't.

   The answer is **D**.

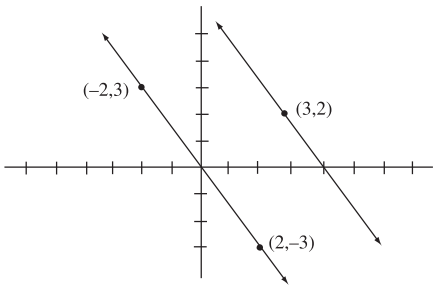


4. A. The formula for the slope is $\frac{y_2 - y_1}{x_2 - x_1}$; but before using it, look at the question again. Since the y-coordinates are equal, the numerator, and thus the fraction, equals 0.

5. E. The slope is equal to

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{a-c}{3c-(-c)} = \frac{a-c}{4c} = 1 \Rightarrow$$
$$a - c = 4c \Rightarrow a = 5c.$$

6. A. Use TACTIC 1: draw a diagram. Quickly sketch the line through (–2, 3) and (2, –3) and the line parallel to it that goes through (3, 2).



Clearly, the slopes of both lines are negative. The answer must be A or B. By KEY FACT N4, the slope of the line through (–2, 3) and (2, –3) is

$$\frac{-3-3}{2-(-2)} = \frac{-6}{4} = -\frac{3}{2}.$$

By KEY FACT N6, nonvertical parallel lines have equal slopes, so the answer is $-\frac{3}{2}$. Note that it is irrelevant that the second line passes through (3, 2).

7. C. By KEY FACT N4, the slope of the line through (–2, 3) and (2, –3) is

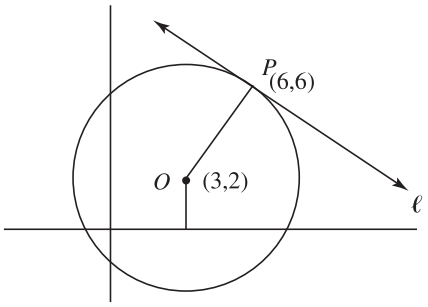
$$\frac{-3-3}{2-(-2)} = \frac{-6}{4} = -\frac{3}{2}.$$

By KEY FACT N6, if two nonvertical lines are perpendicular, the product of their slopes is –1. Then, if m is the slope of the perpendicular line,

$$-\frac{3}{2}m = -1 \Rightarrow -3m = -2 \Rightarrow m = \frac{2}{3}.$$

As in Exercise 6, it is irrelevant that the line passes through (3, 2).

8. B. Draw a rough sketch.



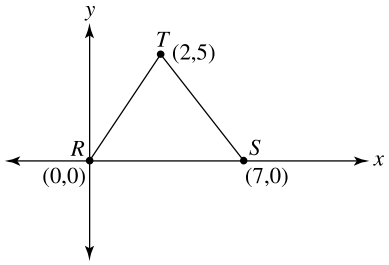
Line segment $\overline{OP}$, joining (3, 2) and (6, –6) is a radius and so, by KEY FACT L10, is perpendicular to line ℓ. The slope of $\overline{OP}$ is $\frac{6-2}{6-3} = \frac{4}{3}$. Therefore, the slope of ℓ is $-\frac{3}{4}$.

9. A. A quick sketch shows that the line that passes through (4, –4) and (4, 4) is vertical. Then, by KEY FACT N7, its equation is $x = 4$.

10. E. Since the line is neither horizontal nor vertical, its equation has the form $y = mx + b$. Since it crosses the y-axis at (0, 5), $b = 5$. Since it passes through (0, 5) and (5, 0), its slope is $\frac{0-5}{5-0} = \frac{-5}{5} = -1$ and its equation is

$$y = -1x + 5 \text{ or } y = -x + 5.$$

11. (1.5) Sketch the triangle, and then calculate the slopes. Since RS is horizontal, its slope is 0.



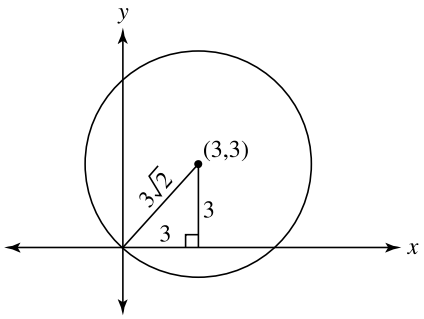
The slope of $RT = \frac{5-0}{2-0} = 2.5$.

The slope of $ST = \frac{5-0}{2-7} = \frac{5}{-5} = -1$.

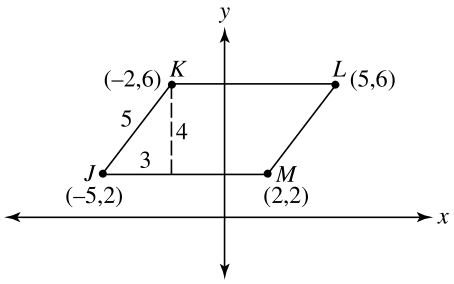
Now add: 0 + 2.5 + (–1) = 1.5.

12. (18) Since the line segment joining (3, 3) and (0, 0) is a radius of the circle, the radius equals $3\sqrt{2}$. Therefore,

$$\text{area} = \pi\left(3\sqrt{2}\right)^2 = 18\pi \Rightarrow k = 18.$$



Here is the diagram for solutions 13 and 14.



13. (28) The base is 7, and the height is 4. The area is $7 \times 4 = 28$.

14. (24) Sides JM and KL are each 7. Also, sides JK and LM are each the hypotenuse of a 3-4-5 right triangle, and so they are 5. The perimeter is $2(7 + 5) = 24$.

15. $\left(38.5 \text{ or } \frac{77}{2}\right)$ Draw in line segment BE, dividing quadrilateral $ABCD$ into rectangle $ABED$ and $\triangle BEC$.

The area of the rectangle is $4 \times 7 = 28$, and the area of the triangle is $\frac{1}{2}(7)(3) = 10.5$. The total area is 38.5 or $\frac{77}{2}$.